

VUB, UNIVERSITY OF ANTWERP

PARTICLE PHYSICS

ANALYSIS NOTE FOR $H + c$ WITH $H \rightarrow WW$

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AUGUST 21, 2026



Contents

1	Datasets	4
1.1	2022 and 2023	4
1.1.1	Data samples	4
1.1.2	Background samples - mostly inclusive samples and no extension samples	5
1.1.3	Background samples - mostly inclusive samples and with extension samples	7
1.1.4	Background samples mostly non-inclusive samples and with extension samples	8
2	Object Definitions	8
2.1	Muons	8
2.2	Electrons	9
3	Efficiencies	10
3.1	New HWW WPs	10
3.1.1	Muon object	10
3.1.2	Electron object	11
3.2	Muon Efficiencies	12
4	Corrections and Weights	15
5	Base selection	15
5.1	Selection based on Run 2 $H(WW) + c$ analysis	15
5.2	Selection based main HWW analysis lepton working points	17
5.3	Results for mostly inclusive samples and no extension samples	20
5.4	Results for mostly inclusive samples and with extension samples	20
5.5	Results for mostly non-inclusive samples and with extension samples	20
5.6	Comparison of inclusive sample selection vs noninclusive	20
6	Top Control Region	20
6.1	Results for mostly inclusive samples and no extension samples	21
6.2	Results for mostly inclusive samples and with extension samples	21
6.3	Results for mostly non-inclusive samples and with extension samples	21
7	High Mass Control Region	21
7.1	Results for mostly inclusive samples and no extension samples	22
7.2	Results for mostly inclusive samples and with extension samples	22
7.3	Results for mostly non-inclusive samples and with extension samples	22
8	MVA Variables	23

9 Feature importance	26
10 Signal Sample Production	27
10.1 Generation	27
10.2 Production chain	27
10.3 Produced samples	28
10.4 Validation	28
10.5 Access	28
10.6 Cross section	28
11 MVA Training	30
11.1 Training dataset	30
11.2 Input variables	31
11.3 Use of the training events in the fit	32
11.4 Model architecture	32
11.5 Six-class training (v11)	33
11.6 κ -HCE training (v32)	34
11.7 Choice of training for the fit	36
12 The κ-HCE Loss	37
12.1 Notation	37
12.2 Dynamic groups from cosine similarity	37
12.3 The three terms	38
12.3.1 Group term	38
12.3.2 Fine term	38
12.3.3 Significance term	39
12.4 Implementation	39
12.5 Status of this loss in the analysis	39
13 Charm Tagging	40
13.1 The two-dimensional scheme	40
13.2 Occupancy in this analysis	40
13.3 Application of the scale factors	41
13.4 Effect on the limit	42
13.5 Use in the MVA	43
14 Negative-Weight Reweighting for V+jets	44
14.1 Origin and consequence of negative weights	44
14.2 The reweighting identity	44
14.3 Variance reduction	45
14.4 Estimation of P_+	45
14.5 Training region	47
14.6 Closure and renormalisation	47
14.7 Uncertainty	49

14.8	Applicability to the jet-binned samples	49
14.9	Limitations of the uncertainty estimate	50
14.10	Relation to existing work	50
15	W+jets Sample Choice	51
15.1	Motivation	51
15.2	Sample replacement	51
15.3	Effective statistics	51
15.4	Effect on the fit	52
16	Statistical Model	54
16.1	Categories	54
16.2	The fit discriminant	56
16.3	Comparison with data	56
16.4	Datacard structure	57
16.5	Normalisation	57
16.6	Object-shift systematics	58
17	Systematic Uncertainties	59
17.1	Shape uncertainties	59
17.2	Treatment of the $t\bar{t}$ theory uncertainties	63
17.3	Rate uncertainties	63
17.4	Top-quark p_T reweighting	64
17.5	Higgs heavy-flavour composition	64
17.6	Missing transverse momentum	64
18	Results	65
18.1	Expected limit	65
18.2	Development of the limit	66
18.3	Impact of individual nuisances	66
18.4	Fit diagnostics	68
18.5	Comparison with the Run 2 analysis	70

Changelog

Below you can find the changes that have been made to this analysis note and on which date + by whom.

- 2026/07/01 - Thomas: added changelog
- 2026/07/28 - Thomas: correction of dz and dxy cuts in electron ID
- 2026/08/03 - Thomas: info on promptMVA working points regarding retraining fro run 3

1 Datasets

1.1 2022 and 2023

1.1.1 Data samples

Table 1: Primary datasets and integrated luminosities for the full 2022 data taking.

Campaign	DAS Path	Luminosity [1/fb]
2022preEE	Muon/Run2022C-22Sep2023-v1/NANOAOD	5.0104
	DoubleMuon/Run2022C-22Sep2023-v1/NANOAOD	
	SingleMuon/Run2022C-22Sep2023-v1/NANOAOD	
	EGamma/Run2022C-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022C-22Sep2023-v1/NANOAOD	
	Muon/Run2022D-22Sep2023-v1/NANOAOD	2.9700
	EGamma/Run2022D-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022D-22Sep2023-v1/NANOAOD	
2022postEE	Muon/Run2022E-22Sep2023-v1/NANOAOD	5.8070
	EGamma/Run2022E-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022E-22Sep2023-v1/NANOAOD	
	Muon/Run2022F-22Sep2023-v2/NANOAOD	17.7819
	EGamma/Run2022F-22Sep2023-v1/NANOAOD	
	MuonEG/Run2022F-22Sep2023-v1/NANOAOD	
	Muon/Run2022G-22Sep2023-v1/NANOAOD	3.0828
	EGamma/Run2022G-22Sep2023-v2/NANOAOD	
	MuonEG/Run2022G-22Sep2023-v1/NANOAOD	

Table 2: Primary datasets and integrated luminosities for the full 2023 data taking.

Campaign	DAS Path	Luminosity [1/fb]
2023preBPix	MuonEG/Run2023C-22Sep2023_v1-v1/NANOAOD	17.7940
	MuonEG/Run2023C-22Sep2023_v2-v1/NANOAOD	
	MuonEG/Run2023C-22Sep2023_v3-v1/NANOAOD	
	MuonEG/Run2023C-22Sep2023_v4-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v1-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v2-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v3-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v4-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v1-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v2-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v3-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v4-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v1-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v2-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v3-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v4-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v1-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v2-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v3-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v4-v2/NANOAOD	
2023postBPix	MuonEG/Run2023D-22Sep2023_v1-v1/NANOAOD	9.4500
	MuonEG/Run2023D-22Sep2023_v2-v1/NANOAOD	
	EGamma0/Run2023D-22Sep2023_v1-v1/NANOAOD	
	EGamma0/Run2023D-22Sep2023_v2-v1/NANOAOD	
	EGamma1/Run2023D-22Sep2023_v1-v1/NANOAOD	
	EGamma1/Run2023D-22Sep2023_v2-v1/NANOAOD	
	Muon0/Run2023D-22Sep2023_v1-v1/NANOAOD	
	Muon0/Run2023D-22Sep2023_v2-v1/NANOAOD	
	Muon1/Run2023D-22Sep2023_v1-v1/NANOAOD	
	Muon1/Run2023D-22Sep2023_v2-v1/NANOAOD	

1.1.2 Background samples - mostly inclusive samples and no extension samples

For two samples, *TBbarQ t-channel 4FS TuneCP5 13p6TeV powheg-madspin-pythia* and *TbarBQ t-channel 4FS TuneCP5 13p6TeV powheg-madspin-pythia8*, we set the cross section to zero in the weighting, as we did not trust the found values. Their contribution is not large, so not including them does not have a huge impact. In fact, in the Run 2 analysis, they do not seem to have been included at all.

1.1 2022 and 2023

Table 3: Non-Higgs background MC samples and cross sections values for the full 2022 and 2023 years.

Process	DAS Path	$\sigma \times \text{BR}$ [pb]	Source
tt			
$tt \rightarrow 2W + 2b \rightarrow 2\ell 2\nu + 2b$	TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	98.578	Twiki for NNLO σ , arxiv:2201.07861 for BR
$tt \rightarrow 2W + 2b \rightarrow (\ell\nu + 2q) + 2b$	TTtoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8	406.263	Twiki for NNLO σ , arxiv:2201.07861 for BR
$tt \rightarrow 2W + 2b \rightarrow 4q + 2b$	TTto4Q_TuneCP5_13p6TeV_powheg-pythia8	418.574	Twiki for NNLO σ , arxiv:2201.07861 for BR
Single Top			
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow 2q + b + 2q$	TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8	19.918	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow \ell\nu + b + 2q$	TWminustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8	19.332	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow \ell\nu + b + \ell\nu$	TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	4.691	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow 2q + \bar{b} + 2q$	TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8	19.918	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow \ell\nu + \bar{b} + 2q$	TbarWplustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8	19.332	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow \ell\nu + \bar{b} + \ell\nu$	TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	4.691	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + q \rightarrow (W^- + \bar{b}) + q \rightarrow 2q + \bar{b} + q$	TbarQtto2Q-t-channel_TuneCP5_13p6TeV_powheg-pythia8	58.703	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + q \rightarrow (W^- + \bar{b}) + q \rightarrow \ell\nu + \bar{b} + q$	TbarQttoLNU-t-channel_TuneCP5_13p6TeV_powheg-pythia8	28.488	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + \bar{q} \rightarrow (W^+ + b) + \bar{q} \rightarrow 2q + b + \bar{q}$	TQbarto2Q-t-channel_TuneCP5_13p6TeV_powheg-pythia8	97.614	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + \bar{q} \rightarrow (W^+ + b) + \bar{q} \rightarrow \ell\nu + b + \bar{q}$	TQbartoLNU-t-channel_TuneCP5_13p6TeV_powheg-pythia8	47.372	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + b \rightarrow (W^- + \bar{b}) + b \rightarrow \ell\nu + \bar{b} + b$	TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8	1.481	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + \bar{b} \rightarrow (W^+ + b) + \bar{b} \rightarrow \ell\nu + b + \bar{b}$	TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8	2.367	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + \bar{b} + q \rightarrow \dots$	TBbarQt-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8	0.000	CMS AN-24-078 says 145.00 pb ??
$\bar{t} + b + q \rightarrow \dots$	TbarBQt-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8	0.000	CMS AN-24-078 says 87.20 pb ??
V + Jets			
$W + 0, 1, 2 \text{ jets} \rightarrow \ell\nu + \text{jets}$	WtoLNU-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	67710.000	XSDB
Diboson			
$ZZ \rightarrow \dots$	ZZ_TuneCP5_13p6TeV_powheg-pythia8	16.70	CMS AN-24-078
$WZ \rightarrow \dots$	WZ_TuneCP5_13p6TeV_powheg-pythia8	54.30	CMS AN-24-078
$WW \rightarrow \dots$	WW_TuneCP5_13p6TeV_powheg-pythia8	122.30	CMS AN-24-078
WG (Diboson)			
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUg-1Jets_PTG-10to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	662.2	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUg-1Jets_PTG-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	2.221	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUg-1Jets_PTG-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	0.2908	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUg-1Jets_PTG-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	0.02231	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1 \text{ jets}$	WGtoLNUg-1Jets_PTG-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	0.004907	XSDB
DY + Jets			
$q\bar{q} \rightarrow Z/\gamma^* + 0, 1, 2 \text{ jets} \rightarrow 2\ell + \text{jets}$	DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	6688.0	XSDB
$q\bar{q} \rightarrow Z/\gamma^* + 0, 1, 2 \text{ jets} \rightarrow 2\ell + \text{jets}$	DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8	20950.0	XSDB

Table 4: Higgs background MC samples (+ private signal MC sample) and cross sections values for both 2022 and 2023

Process	DAS Path	$\sigma \times \text{BR}$ [pb]	Source
Higgs Production			
$gg \rightarrow H \rightarrow WW \rightarrow 2\ell 2\nu$	GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8	1.0825	YR4 SM for σ , arxiv:2201.07861 for BR
$q\bar{q} + VV \rightarrow q\bar{q} + H \rightarrow q\bar{q} + WW \rightarrow q\bar{q} + 2\ell 2\nu$	VBHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8	0.09305	YR4 SM for σ , arxiv:2201.07861 for BR
$q\bar{q} \rightarrow Z + H \rightarrow \dots + WW \rightarrow \dots + 2\ell 2\nu$	ZH_ZtoA11_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-HZJ-jhugenv752-pythia8	0.021559	YR4 SM for σ , arxiv:2201.07861 for BR
$gg \rightarrow Z + H \rightarrow \dots + WW \rightarrow \dots + 2\ell 2\nu$	GluGluZH_ZtoA11_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugenv752-pythia8	0.003094	YR4 SM for σ , arxiv:2201.07861 for BR
$gg \rightarrow H \rightarrow 2\tau$	GluGluHto2Tau_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia	2.976691	YR4 SM for σ , arxiv:2201.07861 for BR
$q\bar{q} + VV \rightarrow q\bar{q} + H \rightarrow 2\tau$	VBHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.255873	YR4 SM for σ , arxiv:2201.07861 for BR
$W^+ + H \rightarrow \ell\nu + WW \rightarrow \ell\nu + 2\ell 2\nu$	WplusH_WtoLNU_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-jhugen-pythia8	0.006625	YR4 SM for σ , arxiv:2201.07861 for BR
$W^- + H \rightarrow \ell\nu + WW \rightarrow q\bar{q} + 2\ell 2\nu$	WminusH_WtoLNU_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-jhugen-pythia8	0.008719	YR4 SM for σ , arxiv:2201.07861 for BR
$W^- + H \rightarrow \dots + 2\tau$	WminusHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8	0.035615	YR4 SM for σ , arxiv:2201.07861 for BR
$W^+ + H \rightarrow \dots + 2\tau$	WplusHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8	0.055763	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow \dots + 2c$	ttHto2C_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.016509	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow \dots + 2\gamma$	ttHtoGG_M-125_TuneCP5_13p6TeV_amcatnloFXFX-madspin-pythia8	0.001296	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow \dots + (Z + \gamma) \rightarrow \dots + \dots + \gamma$	ttHtoZG_ZtoA11_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.000875	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow \dots + 2b$	ttHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8	0.332585	YR4 SM for σ , arxiv:2201.07861 for BR
$H + b \rightarrow WW + b \rightarrow 2\ell 2\nu + b$	private sample	0.002214	
$H + c \rightarrow WW + b \rightarrow 2\ell 2\nu + b$	private sample	0.016972	

Table 5: Missing MC samples for both 2022 and 2023

Process	Sample(s)	Status / Remarks
$H + c$	HPlusCharm	In central production
$H + b$	HPlusBottom	In central production
$W + H \rightarrow \ell\nu + WW \rightarrow \ell\nu + 2\ell 2\nu$	WplusH.WtoLNu.Hto2Wto2L2Nu, WminusH.WtoLNu.Hto2Wto2L2Nu	WplusH available for 2022postEE, not for other years
$W + H \rightarrow qq + WW \rightarrow qq + 2\ell 2\nu$	WplusH.Wto2Q.Hto2Wto2L2Nu, WminusH.Wto2Q.Hto2Wto2L2Nu	WminusH available for 2022postEE, not for other years
$tt + H \rightarrow \dots + \text{non-bb}$	ttHToNonbb	Other non-bb samples, besides $2c, 2\gamma, Z\gamma$ needed?
$Z + H \rightarrow \dots + 2\tau$	ZHToTauTau / ZHTo2Tau	Now available, but not yet used
$tt + H \rightarrow \dots + 2\tau$	ttHToTauTau / ttHTo2Tau	Not available for all Run 3 years

1.1.3 Background samples - mostly inclusive samples and with extension samples

For adding the extension samples, we need to take into account that the number of events increases, but that the total cross section for both the non-extension and the extension sample needs to stay the same as for only using the non-extension sample. Therefore, for the individual samples, we need to weigh the cross section by the number of events in the sample. This is done according to the following scheme. Let N_1 and N_2 be the number of events for the individual samples. $N = N_1 + N_2$ is the total number of events for both samples. σ is the total cross section, which was used when not including any extension sample. Now, let σ_1 and σ_2 be the cross section values for the individual samples. Then we have

$$\sigma_1 = \frac{N_1}{N} \sigma \quad (1.1)$$

$$\sigma_2 = \frac{N_2}{N} \sigma, \quad (1.2)$$

while

$$\sigma = \sigma_1 + \sigma_2 \quad (1.3)$$

The number of events and weighted cross sections for the samples with their extension sample are listed [here](#).

The samples added with respect to the previous section are listed below in blue. These are exclusively extension samples and limit themselves to tt and Single Top processes. The samples that were already included previously are in black.

Table 6: Background MC samples for which extension samples were added with their cross sections values, for both 2022postEE and 2022preEE

Process	DAS Path	$\sigma \times \text{BR}$ [pb]	Source
tt			
$tt \rightarrow 2W + 2b \rightarrow 2\ell 2\nu + 2b$	TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	49.012	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/*ext*	49.567	Twiki for NNLO σ , arxiv:2201.07861 for BR
$tt \rightarrow 2W + 2b \rightarrow (\ell\nu + 2q) + 2b$	TTtoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8	159.559	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TTtoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/*ext*	246.704	Twiki for NNLO σ , arxiv:2201.07861 for BR
$tt \rightarrow 2W + 2b \rightarrow 4q + 2b$	TTto4Q_TuneCP5_13p6TeV_powheg-pythia8	212.832	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/*ext*	205.742	Twiki for NNLO σ , arxiv:2201.07861 for BR
Single Top			
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow 2q + b + 2q$	TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8	9.963	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/*ext*	9.955	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow \ell\nu + b + 2q$	TWminustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8	9.407	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TWminustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/*ext*	9.925	Twiki for NNLO σ , arxiv:2201.07861 for BR
$t + W^- \rightarrow (W^+ + b) + W^- \rightarrow \ell\nu + b + \ell\nu$	TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	2.283	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/*ext*	2.408	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow 2q + \bar{b} + 2q$	TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8	9.655	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/*ext*	10.263	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow \ell\nu + \bar{b} + 2q$	TbarWplustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8	9.158	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TbarWplustoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/*ext*	10.174	Twiki for NNLO σ , arxiv:2201.07861 for BR
$\bar{t} + W^+ \rightarrow (W^- + \bar{b}) + W^+ \rightarrow \ell\nu + \bar{b} + \ell\nu$	TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8	2.280	Twiki for NNLO σ , arxiv:2201.07861 for BR
	TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/*ext*	2.411	Twiki for NNLO σ , arxiv:2201.07861 for BR

1.1.4 Background samples mostly non-inclusive samples and with extension samples

The samples that are now different are most diboson samples: for WW , WZ and ZZ .

2 Object Definitions

2.1 Muons

Main source: <https://muon-wiki.docs.cern.ch/guidelines/recommendations/>

The muon object in nanoAOD is already defined on a few selections. These are applied to the so-called 'reconstructed muon' object (recoMu), which we will call a 'candidate muon'. A reconstructed muon is a track that can be associated to a muon due to its specific signature in the CMS detector, like being detected by the muon chambers. The selections on the candidate muon to define the muon object in nanoAOD are defined as:

```
pt > 15 || (pt > 3 && (passed('CutBasedIdLoose')
|| passed('SoftCutBasedId') || passed('SoftMvaId')
|| passed('CutBasedIdGlobalHighPt') || passed('CutBasedIdTrkHighPt'))))
```

This means that a muon in nanoAOD already has a minimal cut on transverse momentum (p_T) of 3 GeV. For any muon candidate that does not satisfy any of the ID selections, a transverse momentum higher than 15 GeV is required. On top of that, for all muon candidates with $3 \text{ GeV} < p_T \leq 15 \text{ GeV}$, at least one ID selection needs be fulfilled. The definitions of those IDs can be found in Table 7.

Table 7: *Explanation, content and/or selections used to define all muon IDs used in the nanoAOD muon object definition.*

Muon ID Name	Alias	Explanation/Content/Selections
CutBasedIdLoose	Loose ID	
SoftCutBasedId		
SoftMvaId		
CutBasedIdGlobalHighPt		
CutBasedIdTrkHighPt		

We are now converging to the lepton working points used in the WW and HWW analyses. For the muons, the working points show a lot of similarities.

In the main HWW analysis code (Latinos) they use $\text{Muon_pfIsoId} \geq 4$, which means that all standard POG ID's starting from tight or tighter are allowed. $4 = \text{PFIsoTight}$, $5 = \text{PFIsoVeryTight}$, $6 = \text{PFIsoVeryVeryTight}$. This is the `pfIsoId` from miniAOD, where Tight corresponds to `PFIso` below 0.15. in the main HWW analysis note for lepton objects and efficiencies / scale factor they say they use `tightIso`. This ISO is defined as on the muon POG website and tight is then $\text{Iso} < 0.15$. In our framework, we use `Muon.pfRelIso04_all` below 0.15 (if `Muon.pfRelIso04_all` exists). This `pfRelIso04_all` is according to NanoAODv12 documentation (https://cms-xpog.docs.cern.ch/autoDoc/NanoAODv12/2022/2023/doc_EGamma_Run2022F-22Sep2023-v1.html#Muon) PF relative isolation $\text{dR}=0.4$ (deltaBeta corrections). Muon POG website mentions: to access the value of the relative combined PF iso directly, use in nanoAOD `pfRelIso04_all`. So in short, it seems that all those different definitions are actually equivalent.

2.2 Electrons

For the electrons, ID definitions can be found here: <https://twiki.cern.ch/twiki/bin/view/CMS/CutBasedElectronIdentificationRun3>. In the new working point based on the main HWW working points, we use the medium ID working points (for barrel (EB) and endcap (EE)), with an additional tight electron ID on top plus an extra `tthMVA` selection.

The applied ID scale factor is the one for the Medium ID (standard POG definition). for the reco we also apply the standard POG scale factors. On top of those, also a `promptMVA` scale factor is applied that includes the selection with the `tthMVA` and the selection on `dxy` and `dz` (also an additional conversion veto selection is implemented here, although it is already included in the medium ID).

<https://twiki.cern.ch/twiki/bin/view/CMS/MultivariateElectronIdentificationRun3>

<https://twiki.cern.ch/twiki/bin/view/CMS/CutBasedElectronIdentificationRun3>

3 Efficiencies

The data efficiency for Muon TightIso for 2022postEE is lower than for 2022preEE, 2023preBPix and 2023postBPix. The Muon POG has also noticed this and it has to do with neutral isolation mismodeling (see https://indico.cern.ch/event/1336703/contributions/5639886/attachments/2756657/4800108/231121_MuonPOG_PPDWorkhop_wonjun.pdf#page=12, also slide 26 and further, and <https://cms-talk.web.cern.ch/t/muon-tightiso-efficiency-lower-for-2022postee/143784>). The neutral isolation is used in the definition of ParticleFlow isolation (PFIso). And what is observed is a higher Data/MC disagreement of the neutral isolation for 2022postEE. Specifically, there is a problem for the F and G eras. More info here: <https://indico.cern.ch/event/1339886/contributions/5640558/attachments/2741385/4768589/HCAL%20Conditions%20for%202022.pdf>

3.1 New HWW WPs

Currently, the working points for the muon and electron from the main HWW analysis are implemented. Though, an investigation is still ongoing for the electron working point, as the implementation of this new working point caused a massive drop in number of events (data as well as MC). This leads us to believe that the implementation was not done correctly. Therefore, we will eliminate some cuts in the working point, to pinpoint where the drastic cut in events exactly happens, by which selection in the working point.

3.1.1 Muon object

The scale factors for the muon efficiencies are all taken from the main HWW json files. A custom ID is defined, which I call the ‘TightHWW’ ID. This is a selection including the standard Tight ID from the muon POG, as well as the other cuts on the muon object (except for the Tight Iso and promptMVA mvaTTH working points). An overview is given below.

Table 8: Definition of the TightHWW muon ID working point.

Object	Criteria	Selection
Muon	p_T	$> 10 \text{ GeV}$
	$ \eta $	< 2.4
	$ d_z $	$< 0.1 \text{ cm}$
	$ d_{xy} $	$< 0.01 \text{ cm for } p_T < 20 \text{ GeV}$
	$ d_{xy} $	$< 0.02 \text{ cm for } p_T \geq 20 \text{ GeV}$
ID		Tight

The scale factors we apply for the muon are the scale factors privately derived for the custom TightHWW ID, the standard muon POG Tight Iso and the promptMVA working point selection (`mvaTTH` > 0.67). No scale factor on the muon reconstruction is applied, following the muon POG recommendations, as those scale factors are essentially equal to 1.

3.1.2 Electron object

The scale factors for the electron efficiencies are taken from the centrally derived EGamma POG json files for the electron medium ID working point and from the main HWW json files for the promptMVA working point. The promptMVA working point is called ‘tthMVA_Run3’ (another working point ‘tthMVA_HWW’, might be used later, but no fake rates are available for that one yet. The only difference is a distinction for the `mvaTTH` cut for $p_T \leq 20$ GeV). It includes the following cuts:

Table 9: Definition of the *tthMVA_Run3* electron promptMVA working point.

Object	Criteria	Selection
Electron	p_T	> 10 GeV
	$ \eta $	< 2.5
	promptMVA	<code>mvaTTH</code> > 0.90
	conversion veto	<code>convVeto</code> = True
	$ d_z $	< 0.1 cm for $ \eta \leq 1.479$ (EB) < 0.2 cm for $ \eta > 1.479$ (EE)
	$ d_{xy} $	< 0.05 cm for $ \eta \leq 1.479$ (EB) < 0.1 cm for $ \eta > 1.479$ (EE)

The orange coloured cuts in the table above for the scale factors, point out that the *tthMVA_Run3* working point initially does not include those cuts, but in the derivation of the scale factor, they are included in this working point. This is just a matter of how the working points are implemented in the code, but should not make a difference on the final number of events. It is purely to compensate the cut on p_T and $|\eta|$ not being included in the derivation of the medium ID scale factor by the EGamma POG.

Also note that the conversion veto requirement is actually already included in the definition of the medium ID working point from the EGamma POG. It has just been plugged in by the main HWW analysis people as an additional certainty. This poses no problem, as the *tthMVA_Run3* working point scale factors are derived with the medium ID working point in the denominator.

On the twiki (<https://twiki.cern.ch/twiki/bin/view/CMS/CutBasedElectronIdentification>) it is stated that cuts on $|d_z|$ and $|d_{xy}|$ are not included in the standard Cut Based ID. Therefore, these are manually put in as an extra cut. They are also not included in the Scale factors for the Cut Based ID electron WPs from the EGamma POG and

thus need to be included in an additional scale factor derivation for the total electron efficiency. In this analysis, these efficiencies and scale factors are included in the scale factor calculation of the electron promptMVA working point.

The promptMVA working points of HWW actually use a retrained version of the TTH mva score, both for electron and muon. For 2022 and 2023 (nanoAOD version 12), this is NOT implemented yet in nanoAOD. For 2024, with nanoAOD version 15, it is, and is just called `promptMVA`, instead of `mvaTTH`. This could explain why using the `mvaTTH` cut for electrons gives such bad results (cuts a lot of events and worsens the data/MC ratio by about 15 %). I will now try to implement the corrected mva scores. This is through an xml file, containing the weights from the TTH MVA training. One xml for electron and one for muon.

3.2 Muon Efficiencies

The efficiency calculation of the muon object is divided into four parts: the reconstruction efficiency, the ID efficiency, the Iso efficiency and the promptMVA efficiency. To obtain a full muon efficiency, we can just multiply these efficiencies together:

$$\varepsilon_{\mu} = \varepsilon_{\text{reco}} \cdot \varepsilon_{\text{ID}} \cdot \varepsilon_{\text{Iso}} \cdot \varepsilon_{\text{promptMVA}} \quad (3.1)$$

The same can be done for the muon scale factors. The only requirement to factorize the efficiencies and scale factors this way is that the denominators match the other factors in the factorization. An example is that the reco selection used in the numerator of the first factor must appear in the denominator of the second factor. Then both the reco and ID selection should be used in the denominator of the third factor, and so on. . Thus, for our case, the full muon efficiency, without trigger efficiencies becomes:

$$\begin{aligned} \varepsilon_{\mu} = & \varepsilon_{\text{Tracker Muon}} \times \varepsilon_{\text{TightHWW ID} \mid \text{Tracker Muon}} \\ & \times \varepsilon_{\text{Tight Iso} \mid \text{TightHWW ID \& Tracker Muon}} \\ & \times \varepsilon_{\text{promptMVA} \mid \text{Tight Iso \& TightHWW ID \& Tracker Muon}} \end{aligned} \quad (3.2)$$

The trigger efficiencies need to be more delicately combined, as we use not only single muon efficiencies.

The Tracker POG has calculated the tracker reconstruction scale factors for $p_T > 20 \text{ GeV}$ **what about $p_T < 20 \text{ GeV}$?** and concluded that the scale factors are all very close **how close?** to 1 **add ref:** <https://muon-wiki.docs.cern.ch/guidelines/corrections/> (for Run 2 available here, for Run 3 asked muon POG privately). Hence, the Muon POG has recommended to neglect this effect, which we do here. This simplifies our formula to

$$\begin{aligned} \varepsilon_{\mu} = & \varepsilon_{\text{TightHWW ID} \mid \text{Tracker Muon}} \times \varepsilon_{\text{Tight Iso} \mid \text{TightHWW ID \& Tracker Muon}} \\ & \times \varepsilon_{\text{promptMVA} \mid \text{Tight Iso \& TightHWW ID \& Tracker Muon}} \end{aligned} \quad (3.3)$$

The remaining muon efficiencies in the factorized formula above are derived using the Tag and Probe method [add ref:N. Adam et al., “Generic tag and probe tool for measuring efficiency at cms with early data”, CMS Note 2009/111 \(2009\).](#) We use the official Spark_TnP package [add ref: <https://twiki.cern.ch/twiki/bin/viewauth/CMS/MuonTagAndProbeTreesRun2>](#) from the Muon POG for this.

This method relies on selecting a Tag with a very tight selection, optimizing the probability for it to be a real muon. Pairs of the Tag with a Probe are then formed for Drell-Yan events, such that we know that when we have a pair coming from a Z and one of the objects in the pair is a muon (the Tag), the other (the Probe) must be a muon as well. On this Probe we then test our efficiencies as it can then be selected very loosely, such that the efficiency calculation is not biased. Below, you can find the different selections applied to the Tag. The Probe always needs to contain the actual selection we want to calculate the efficiency of and therefore varies.

Table 10: Definition of the Tag selection for muon efficiency calculations.

Object	Criteria	Selection
Tag Muon	p_T	$> 26 \text{ GeV}$
	$ \eta $	< 2.4
	$ d_z $	$< 0.5 \text{ cm}$
	$ d_{xy} $	$< 0.2 \text{ cm}$
	ID	Tight
	Iso	< 0.2
	Trigger	HLT_IsoMu24

The Iso is the standard Muon Particle-Flow Isolation from the Muon POG [<https://muon-wiki.docs.cern.ch/guidelines/recommendations/#particle-flow-isolation>]. As we use it in the Spark_TnP package at different values for systematics estimates, it is explicitly defined in our Spark_TnP config file as:

$$Iso^\mu = \frac{\sum_{\text{ch. had from PV } i} p_T^i + \max \left[0, \sum_{\text{neut. had } j} E_T^j + \sum_{\gamma} E_T^\gamma - 0.5 \sum_{\text{ch. had from PU } k} p_T^k \right]}{p_T^\mu} \quad (3.4)$$

For the TightHWW ID efficiency, we run with the TightHWW selection in the numerator, Tracker Muons in the denominator, and no additional selection on the probe except for this demand:

```
pair_probeMultiplicity and (!probe_isDuplicated or probe_isBestPair)
```

following the official Muon POG Run 3 config file for Spark_TnP using muon pairs coming from a Z: https://gitlab.cern.ch/cms-muonPOG/spark_tnp/-/blob/Automatization/

[TagAndProbeCode/configs/muon_pog_official_nanoRun3_Z.json?ref_type=heads](#)

The Datasets we use are the standard datasets from the Muon POG already available in the Spark_TnP package: https://gitlab.cern.ch/cms-muonPOG/spark_tnp/-/blob/Automatization/TagAndProbeCode/data/registry_muon_Z_generalTracks.json?ref_type=heads (in those datasets it says 'Oct23' for their parquet files: see what is really the original MiniAOD dataset. Is it 'Sep23'? - > does main HWW use custom dataset?

Here you can see what datasets are used when making the Ntuples yourself: https://gitlab.cern.ch/cms-muonPOG/spark_tnp/-/blob/Automatization/Workflows/Muon_TnP_cfg.py?ref_type=heads, which is used in this: https://gitlab.cern.ch/cms-muonPOG/spark_tnp/-/blob/7bcadeb40b19c8481a33c2d8ad0a0db225512d59/TagAndProbeNtuples/python/TNP_Muon_POG.py. Typically, we don't need this as you can read here: https://gitlab.cern.ch/cms-muonPOG/spark_tnp/-/blob/0871e289a6ace179348acb41e4cd30a8f3397fa3/README.md.

Here you can find the original central MiniAOD datasets that were used to create the MUOPOG NanoAOD files: https://gitlab.cern.ch/cms-muonPOG/spark_tnp/-/blob/Automatization/TagAndProbeCode/scripts/MUONanoProducer/doHTCondor.py. Finally these MUOPOG Nano files are converted into parquet files using this https://gitlab.cern.ch/cms-muonPOG/spark_tnp/-/blob/Automatization/Workflows/tnp_ntuples.py?ref_type=heads. These parquet files are the ones used in the spark_TnP package to derive muon efficiencies. I can confirm that actual datasets used for data are in fact the 22Sep2023 ones. The name Oct23 used in the data registry of Spark_TnP probably refers to date on which the Muon POG generated the MUOPOG format datasets from these original 22Sep2023 datasets.

Table 11: Samples used for deriving the 2022 and 2023 muon efficiencies.

Campaign	Sample	Run and reconstruction version
2022preEE	SingleMuon	Run2022B-22Sep2023-v1
	SingleMuon	Run2022C-22Sep2023-v1
	Muon	Run2022C-22Sep2023-v1
	Muon	Run2022D-22Sep2023-v1
	Monte-Carlo	DYto2L-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8
2022postEE	Muon	Run2022E-22Sep2023-v1
	Muon	Run2022F-22Sep2023-v1
	Muon	Run2022G-22Sep2023-v1
	Monte-Carlo	DYto2L-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8
2023preBPix	Muon0	Run2023C-22Sep2023_v1-v1
	Muon0	Run2023C-22Sep2023_v2-v1
	Muon0	Run2023C-22Sep2023_v3-v1
	Muon0	Run2023C-22Sep2023_v4-v1
	Muon1	Run2023C-22Sep2023_v1-v1
	Muon1	Run2023C-22Sep2023_v2-v1
	Muon1	Run2023C-22Sep2023_v3-v1
	Muon1	Run2023C-22Sep2023_v4-v2
	Monte-Carlo	DYto2L-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8
2023postBPix	Muon0	Run2023D-22Sep2023_v1-v1
	Muon0	Run2023D-22Sep2023_v2-v1
	Muon1	Run2023D-22Sep2023_v1-v1
	Muon1	Run2023D-22Sep2023_v2-v1
	Monte-Carlo	DYto2L-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8

4 Corrections and Weights

In this section, we describe which corrections and weights are applied to the events and objects in the H(WW)+c analysis framework.

5 Base selection

Selection to verify the control regions are defined in a logical manner. This selection also allows us to see whether different combinations of samples have the same overall shape of the distributions. This is to verify that the stitching of noninclusive samples is done correctly and that the values for the cross sections are correct.

5.1 Selection based on Run 2 $H(WW) + c$ analysis

The event and object selection that was based on the one from Run 2 can be found in the table below.

5.1 Selection based on Run 2 $H(WW) + c$ analysis

Table 12: Object selection criteria based on Run 2 $H(WW) + c$ analysis.

Object	Criteria	Selection
Muon	p_T	$> 10 \text{ GeV}$
	$ \eta $	< 2.4
	ID	Tight
	Iso	Tight
Electron	p_T	$> 10 \text{ GeV}$
	$ \eta $	< 2.5
	ID	mvaIso_WP80
	$\Delta R(e, \mu)$	> 0.4
Jet	p_T	$> 20 \text{ GeV}$
	$ \eta $	< 2.5
	jetId	= 6 (tightLepVeto)
	$\Delta R(J, e/\mu)$	> 0.4
c-jet	is a Jet	
		ParticleNet b-tagger medium wp
		2022 preEE: $CvB > 0.306$ AND $CvL > 0.160$
	c -tagged	2022 postEE: $CvB > 0.304$ AND $CvL > 0.160$
		2023 preBPix: $CvB > 0.353$ AND $CvL > 0.148$
		2023 postBPix: $CvB > 0.157$ AND $CvL > 0.109$
Lepton Pair	$p_T(\ell_1)$	$> 20 \text{ GeV}$
	$p_T(\ell_2)$	$> 10 \text{ GeV}$
	$q(\ell_1) + q(\ell_2)$	= 0
	$m_{\ell\ell}$	$> 12 \text{ GeV}$
	$p_T^{\ell\ell}$	$> 30 \text{ GeV}$

5.2 Selection based main HWW analysis lepton working points

Table 13: Event selection criteria based on Run 2 $H(WW) + c$ analysis.

Criteria	Selection
Trigger	SingleMu SingleEle MuEle
MET filters	see below
nPV	> 0
$N(\ell\ell \text{ pair})$	$= 1$
one e , one μ	$\ell\ell$ pair with lepton 1 is e and lepton 2 is μ or vice versa
p_T^{miss} (Puppi MET)	$> 45 \text{ GeV}$

To **avoid duplicate events** from different primary datasets, events are selected as follows:

- from **EGamma** if the event **passes SingleEle**,
- from **Muon** if the event **passes SingleMu**, while **failing SingleEle**,
- from **MuonEG** if the event **passes MuEle** triggers, while **failing SingleEle and SingleMu**.

Table 14: HLT trigger paths and their corresponding defined flags for Run 3, based on Run 2 $H(WW) + c$ analysis.

Flag	HLT Paths
SingleMu	IsoMu24
SingleEle	Ele30_WPTight_Gsf
MuEle	Mu23.TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL Mu12.TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ

5.2 Selection based main HWW analysis lepton working points

Recently, we moved to the lepton working points of the main HWW analysis that uses the Latinos framework (for VBF and ggH analysis). This is the most recent version of the object and event selections. Shown is the base selection, which is the starting point on which the selections for the control regions, signal region and the MVA training dataset are based. For now, the triggers are still the same ones as above, but an additional trigger might be added, to be fully equivalent with the HWW analysis.

5.2 Selection based main HWW analysis lepton working points

Table 15: Object selection criteria based on main HWW analysis lepton WPs.

Object	Criteria	Selection
Muon	p_T	$> 10 \text{ GeV}$
	$ \eta $	< 2.4
	$ d_z $	$< 0.1 \text{ cm}$
	$ d_{xy} $	$< 0.01 \text{ cm for } p_T < 20 \text{ GeV}$ $< 0.02 \text{ cm for } p_T \geq 20 \text{ GeV}$
	ID	Tight
	Iso	Tight
	promptMVA	mvaTTH > 0.67
Electron	p_T	$> 10 \text{ GeV}$
	$ \eta $	< 2.5
	ID	Medium
	promptMVA	mvaTTH > 0.90
	conversion veto	convVeto = True
	$ d_z $	$< 0.1 \text{ cm for } \eta \leq 1.479 \text{ (EB)}$ $< 0.2 \text{ cm for } \eta > 1.479 \text{ (EE)}$
	$ d_{xy} $	$< 0.05 \text{ cm for } \eta \leq 1.479 \text{ (EB)}$ $< 0.1 \text{ cm for } \eta > 1.479 \text{ (EE)}$
	$\Delta R(e, \mu)$	> 0.4
Jet	p_T	$> 20 \text{ GeV}$
	$ \eta $	< 2.5
	jetId	= 6 (tightLepVeto)
	$\Delta R(J, e/\mu)$	> 0.4
c-jet	is a Jet	
		ParticleNet b-tagger medium wp
		2022 preEE: $CvB > 0.306$ AND $CvL > 0.160$
		2022 postEE: $CvB > 0.304$ AND $CvL > 0.160$
		2023 preBPix: $CvB > 0.353$ AND $CvL > 0.148$
	c -tagged	2023 postBPix: $CvB > 0.157$ AND $CvL > 0.109$
Lepton Pair	$p_T(\ell_1)$	$> 20 \text{ GeV}$
	$p_T(\ell_2)$	$> 10 \text{ GeV}$
	$q(\ell_1) + q(\ell_2)$	= 0
	$m_{\ell\ell}$	$> 12 \text{ GeV}$
	$p_T^{\ell\ell}$	$> 30 \text{ GeV}$

5.2 Selection based main HWW analysis lepton working points

Table 16: Event selection criteria based on main HWW analysis lepton WPs.

Criteria	Selection
Trigger	SingleMu SingleEle MuEle
MET filters	see below
nPV	> 0
$N(\ell\ell \text{ pair})$	$= 1$
one e , one μ	$\ell\ell$ pair with lepton 1 is e and lepton 2 is μ or vice versa
p_T^{miss} (Puppi MET)	$> 45 \text{ GeV}$

To **avoid duplicate events** from different primary datasets, events are selected as follows:

- from **EGamma** if the event **passes SingleEle**,
- from **Muon** if the event **passes SingleMu**, while **failing SingleEle**,
- from **MuonEG** if the event **passes MuEle** triggers, while **failing SingleEle and SingleMu**.

Table 17: HLT trigger paths and their corresponding defined flags for Run 3, on main HWW analysis lepton WPs.

Flag	HLT Paths
SingleMu	IsoMu24
SingleEle	Ele30_WPTight_Gsf
MuEle	Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL Mu12_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ

The applied MET filters are the ones from the POG Run 3 recommendations and have not changed with converging to the HWW working points.

Table 18: MET filters for Run 3.

MET Filters (POG Recommendation)
goodVertices
globalSuperTightHalo2016Filter
EcalDeadCellTriggerPrimitiveFilter
BadPFMuonFilter
BadPFMuonDzFilter
hfNoisyHitsFilter
eeBadScFilter
ecalBadCalibFilter

5.3 Results for mostly inclusive samples and no extension samples

5.4 Results for mostly inclusive samples and with extension samples

5.5 Results for mostly non-inclusive samples and with extension samples

To get the best possible statistics (highest number of MC events), we make a stitching of noninclusive samples combined with inclusive samples to fill in possible gaps that are not covered by any noninclusive sample.

5.6 Comparison of inclusive sample selection vs noninclusive

Here we verify that the shape of the results for mostly non-inclusive samples is the same as that for the results for mostly inclusive samples. Both are with extension samples, where possible, to get the highest amount of statistics possible. For the mostly inclusive samples we only show the total sum of the samples.

6 Top Control Region

The top control region is defined with these additional cuts on top of the selection in the Base selection. In red, the cuts are shown that are different when comparing to the high mass control region.

Criteria	Selection
$N(\text{c-jet})$	≥ 2
$m_{\ell\ell}$	$> 72 \text{ GeV}$
$m_T^{\ell\ell}$	$\leq 60 \text{ GeV}$
$m_T^{\ell_2}$	$> 30 \text{ GeV}$

6.1 Results for mostly inclusive samples and no extension samples

6.2 Results for mostly inclusive samples and with extension samples

6.3 Results for mostly non-inclusive samples and with extension samples

To get the best possible statistics (highest number of MC events), we make a stitching of noninclusive samples combined with inclusive samples to fill in possible gaps that are not covered by any noninclusive sample.

7 High Mass Control Region

The high mass control region is defined with these additional cuts on top of the selection in the Base selection. In red, the cuts are shown that are different when comparing to the top control region.

Criteria	Selection
$N(\text{c-jet})$	≥ 1
$m_{\ell\ell}$	$> 72 \text{ GeV}$
$m_T^{\ell\ell}$	$> 60 \text{ GeV}$
$m_T^{\ell_2}$	$> 30 \text{ GeV}$

7.1 Results for mostly inclusive samples and no extension samples

7.2 Results for mostly inclusive samples and with extension samples

7.3 Results for mostly non-inclusive samples and with extension samples

To get the best possible statistics (highest number of MC events), we make a stitching of noninclusive samples combined with inclusive samples to fill in possible gaps that are not covered by any noninclusive sample.

8 MVA Variables

For reference, below is the table of the Run 2 AN-23-102:

Table 19: Run 2 summary of variables used in BDT training

Variable	Description
$p_{\text{T}}^{\ell\ell}$	momentum of dilepton system ($e\mu$)
$p_{\text{T}}^{\ell_1}$	momentum of leading lepton (sorted by p_{T})
$p_{\text{T}}^{\ell_2}$	momentum of subleading lepton (sorted by p_{T})
p_{T}	momentum of charm jet candidate
$p_{\text{T}}^{\text{miss}}$	missing transverse momentum
$m_{\text{T}}^{\ell_1}$	transverse mass between leading lepton and $p_{\text{T}}^{\text{miss}}$
$m_{\text{T}}^{\ell_2}$	transverse mass between subleading lepton and $p_{\text{T}}^{\text{miss}}$
$m_{\ell\ell}$	mass of dilepton system ($e\mu$)
$\Delta R(\ell\ell, \ell_1)$	ΔR separation between dilepton system and leading lepton
$\Delta R(\ell\ell, \ell_2)$	ΔR separation between dilepton system and subleading lepton
$\Delta R(\ell\ell, c)$	ΔR separation between dilepton system and charm jet
$\Delta\phi(\ell\ell + MET, c)$	$\Delta\phi$ separation of dilepton+MET(Higgs) system and charm jet
$\Delta\phi(\ell_1, MET)$	$\Delta\phi$ separation of leading lepton and charm jet
$\Delta\phi(\ell_2, MET)$	$\Delta\phi$ separation of subleading lepton and charm jet
DeepJet CvL	charm jet discriminant for suppressing light jet
DeepJet CvB	charm jet discriminant for suppressing b jet
nSV	number of Secondary Vertices in this event

And this is the one we use for Run 3:

Variable	Description
$p_T^{\ell\ell}$	transverse momentum of dilepton system ($e\mu$)
$p_T^{\ell_1}$	transverse momentum of leading lepton (sorted by p_T)
$p_T^{\ell_2}$	transverse momentum of subleading lepton (sorted by p_T)
p_T^c	transverse momentum of charm jet candidate (c-jet candidate)
$p_T^{j_1}$	transverse momentum of leading jet
$p_T^{j_2}$	transverse momentum of subleading jet
$p_T^{j_3}$	transverse momentum of third leading jet
p_T^{miss}	missing transverse momentum
$\phi(\text{MET})$	phi of missing transverse momentum
p_T^{jets}	missing transverse momentum of the jets
$\phi(\text{jets})$	azimuthal angle of the jets
$\eta(\text{jets})$	pseudorapidity of the jets
$\eta(\text{jets})$	pseudorapidity of the jets
$N(\text{jets})$	number of jets - jet multiplicity
$N(\text{c-jets})$	number of c-jets - c-jet multiplicity
$m_T^{\ell_1}$	transverse mass between leading lepton and p_T^{miss}
$m_T^{\ell_2}$	transverse mass between subleading lepton and p_T^{miss}
$m_{\ell\ell}$	mass of dilepton system ($e\mu$)
$\Delta R(\ell\ell, \ell_1)$	ΔR separation between dilepton system and leading lepton
$\Delta R(\ell\ell, \ell_2)$	ΔR separation between dilepton system and subleading lepton
$\Delta R(\ell\ell, c)$	ΔR separation between dilepton system and c-jet candidate
$\Delta R(\ell\ell, j_1)$	ΔR separation between dilepton system and leading jet
$\Delta R(\ell\ell, j_2)$	ΔR separation between dilepton system and subleading jet
$\Delta R(\ell\ell, j_3)$	ΔR separation between dilepton system and third leading jet
$\Delta\phi(\ell\ell + MET, c)$	$\Delta\phi$ separation of dilepton+MET(Higgs) system and c-jet candidate
$\Delta\phi(\ell\ell + MET, j_1)$	$\Delta\phi$ separation of dilepton+MET(Higgs) system and leading jet
$\Delta\phi(\ell\ell + MET, j_2)$	$\Delta\phi$ separation of dilepton+MET(Higgs) system and subleading jet
$\Delta\phi(\ell\ell + MET, j_3)$	$\Delta\phi$ separation of dilepton+MET(Higgs) system and third leading jet
$\Delta\phi(\ell_1 + MET, c)$	$\Delta\phi$ separation of leading lepton+MET(Higgs) system and c-jet candidate
$\Delta\phi(\ell_1 + MET, c)$	$\Delta\phi$ separation of leading lepton+MET(Higgs) system and c-jet candidate
$\Delta\phi(\ell_1 + MET, j_1)$	$\Delta\phi$ separation of leading lepton+MET(Higgs) system and leading jet
$\Delta\phi(\ell_1 + MET, j_2)$	$\Delta\phi$ separation of leading lepton+MET(Higgs) system and subleading jet
$\Delta\phi(\ell_1 + MET, j_3)$	$\Delta\phi$ separation of leading lepton+MET(Higgs) system and third leading jet
$\Delta\phi(\ell_1, \text{MET})$	$\Delta\phi$ separation of leading lepton and MET*
$\Delta\phi(\ell_2, \text{MET})$	$\Delta\phi$ separation of subleading lepton and MET*
$\Delta\phi(\ell\ell, \text{MET})$	$\Delta\phi$ separation of dilepton system and MET
c-jet ParticleNet CvL	charm jet discriminant for suppressing light jet for the c-jet candidate
c-jet ParticleNet CvB	charm jet discriminant for suppressing b jet for the c-jet candidate
leading jet jet ParticleNet CvL	charm jet discriminant for suppressing light jet for the leading jet
leading jet ParticleNet CvB	charm jet discriminant for suppressing b jet for the leading jet
nSV	number of Secondary Vertices

*The Run 2 AN-23-102 does say 'charm jet' here, but the variable in the table and on the plots says 'MET', so I assume it should be 'MET'.

To select the lepton pair, the muons and electrons (**object definition to be added**) are put in one object, called 'leptons'. These leptons are then sorted by transverse momentum (p_T) in descending order, making the first element the lepton with highest p_T . Then a 'dilepton' object is constructed for each event containing at least two leptons. The dilepton object consists of the two leptons with the highest p_T in the event. In a specific event, the 'first lepton' is the lepton with the largest p_T of the leptons in an event. The second lepton is the lepton with the second largest p_T in the event. So, the dilepton object consists of the first and second lepton. The lepton-lepton ($\ell\ell$) pair is then that same dilepton object, with the first lepton labeled as ℓ_1 and the second lepton as ℓ_2 . We then also demand that both leptons have opposite charge, that the p_T of the first lepton ($p_T^{\ell_1}$) is above 20 GeV and that of the second lepton ($p_T^{\ell_2}$) is larger than 10 GeV. On top of that the pair should have an invariant mass $m_{\ell\ell}$ above 12 GeV and a transverse momentum $p_T^{\ell\ell}$ above 30 GeV.

$$m_{\ell\ell} = \sqrt{(p^{\ell_1} + p^{\ell_2})^2} \quad (8.1)$$

$$p_T^{\ell\ell} = |\mathbf{p}_T^{\ell\ell}| \quad (8.2)$$

$$\mathbf{p}_T^{\ell\ell} = \mathbf{p}_T^{\ell_1} + \mathbf{p}_T^{\ell_2} \quad (8.3)$$

where p^{ℓ_1} and p^{ℓ_2} are the four-momentum vectors of the first and second lepton, respectively. In natural units:

$$p^{\ell_1} = (E^{\ell_1}, \mathbf{p}_T^{\ell_1}, p_z^{\ell_1}) \quad (8.4)$$

$$p^{\ell_2} = (E^{\ell_2}, \mathbf{p}_T^{\ell_2}, p_z^{\ell_2}) \quad (8.5)$$

where $\mathbf{p}_T^{\ell_1} = (p_x^{\ell_1}, p_y^{\ell_1})$ and $\mathbf{p}_T^{\ell_2} = (p_x^{\ell_2}, p_y^{\ell_2})$.

The charm jet candidate (c-jet candidate) of an event is taken as the c-jet (**object definition of cjets to be added**) with the highest ParticleNet CvL score (**btagPNetCvL**).

The MET object is just the PuppMET object from NanoAOD. p_T^{miss} is the transverse momentum of the MET object, also called 'missing p_T '. Together with this p_T^{miss} , we construct the transverse masses $m_T^{\ell_1}$, $m_T^{\ell_2}$ and $m_T^{\ell\ell}$:

$$m_T^{\ell_1} = \sqrt{2 \cdot p_T^{\ell_1} \cdot p_T^{\text{miss}} \cdot (1 - \cos[\Delta\phi(\ell_1, \text{MET})])} \quad (8.6)$$

$$m_T^{\ell_2} = \sqrt{2 \cdot p_T^{\ell_2} \cdot p_T^{\text{miss}} \cdot (1 - \cos[\Delta\phi(\ell_2, \text{MET})])} \quad (8.7)$$

$$m_T^{\ell\ell} = \sqrt{2 \cdot p_T^{\ell\ell} \cdot p_T^{\text{miss}} \cdot (1 - \cos[\Delta\phi(\ell\ell, \text{MET})])} \quad (8.8)$$

add definition (formula) of $\Delta\phi$.

ΔR is ... (add formula)

nSV is ... (describe what a secondary vertex is)

9 Feature importance

The feature importance of each of the variables is examined using an α -weighted feature importance score. This is based on the integrated gradients approach for determining what features contribute most in the MVA training.

10 Signal Sample Production

No central production exists for $H + c$ or $H + b$ with $H \rightarrow WW$ at $\sqrt{s} = 13.6$ TeV, so both signal samples were produced privately for all four Run 3 campaigns considered in this note.

10.1 Generation

The samples are generated with MadGraph5_aMC@NLO interfaced to Pythia8 with the CP5 tune, in the four-flavour scheme:

```
HPlusCharm_4FS_MuRFSscaleDynX0p50_M125_TuneCP5_13p6TeV_amcatnloFXFX_pythia8
```

and the corresponding HPlusBottom process. A dynamical renormalisation and factorisation scale with $\mu_{R,F}$ scaled by 0.50 is used, and the Higgs boson mass is set to 125 GeV.

Gridpacks were produced with the `genproductions` scripts:

```
export SCRAM_ARCH=el8_amd64_gcc10
./gridpack_generation.sh NAMEOFGP CARDDIR
```

under CMSSW_12_4_8. The gridpacks must be built on an EL8 architecture; an initial production on lxplus9 had to be repeated on lxplus8, as the architecture is not supported for gridpack generation.

10.2 Production chain

Each campaign is produced through the standard multi-step CMS chain, submitted with CRAB:

Table 20: Production steps. Step 1 takes approximately one day per campaign.

Step	Content
1	wmLHEGS — LHE generation, hadronisation, Geant4 simulation
2	DRPremix — digitisation and pileup premixing
3	RECO — reconstruction
4	MiniAODv4
5	NanoAODv13

The DRPremix step requires more memory than the default grid configuration provides and is submitted with an increased memory request.

10.3 Produced samples

All datasets follow the naming pattern

`/HPlusCharm_HToWW_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/<processing>/USER`

and are published on the `prod/phys03` DBS instance. Event counts are taken from `dasgoclient`.

Table 21: Privately produced signal NanoAOD samples.

Sample	Campaign	Events
$H + c$	2022preEE	328 838
$H + c$	2022postEE	277 345
$H + c$	2023preBPix	280 356
$H + b$	2022preEE	809 802
$H + b$	2022postEE	298 177

The remaining campaigns (2023postBPix for both $H + c$ and $H + b$, and 2023preBPix for $H + b$) are in production.

10.4 Validation

TODO: no validation material for the private samples is included yet. Required: a generator-level comparison against a central sample of the same process where one exists, confirmation that the decay chain is the intended one, and confirmation that the pileup profile and global tag match the corresponding central campaign. Also state the PDF set, the FxFx merging scale and the number of matrix-element jets, none of which are currently given.

10.5 Access

The privately produced samples are hosted at the IIHE Tier-2 and are not served by the global XRootD redirector, so they are read through the site redirector directly.

10.6 Cross section

The signal template is normalised to $\sigma \times \mathcal{B} = 2.214 \times 10^{-3}$ pb for $H \rightarrow WW \rightarrow 2\ell 2\nu$. This value defines the signal-strength parameter $r = 1$ and therefore sets the scale of every limit quoted in Sec. 18.

TODO: give the provenance of this number. Required: the order of the calculation and its source, the $H \rightarrow WW \rightarrow 2\ell 2\nu$ branching fraction used and its reference, whether the cross section is inclusive or carries a charm-quark p_T or η requirement at

generator level, and the value of κ_c assumed. Since r is dimensionless, this convention must also be stated alongside the Run 2 comparison in Sec. 18.

11 MVA Training

The analysis uses a multi-class neural network trained with the **b-hive** framework to separate the $H + c$ signal from the dominant backgrounds. Two training schemes are described here: the six-class flat cross-entropy training (**v11**), which defines the categories used for the final fit, and the thirteen-class κ -HCE training (**v32**), which retains fine-grained Higgs sub-class information.

11.1 Training dataset

The training input is produced by the same processor and base selection described earlier in this note, written in **parquet** format. Events are split 80:20 into training and test sets using a deterministic hash of the NanoAOD **event** identifier, so that the split is reproducible and does not depend on file ordering.

Table 22: Training dataset composition, in the six-class scheme, summed over the three Run 3 campaigns used for training. The signal class is smaller than the dominant $t\bar{t}$ class by a factor 523, which motivates the use of inverse-frequency class weights in the loss. Rows sum to the totals.

Class	Train	Test	Total	Fraction
hplusc	4 860	1 215	6 075	0.15%
higgsbkg	279 546	69 884	349 430	8.57%
tt	2 541 821	635 455	3 177 276	77.88%
st	398 285	99 571	497 856	12.20%
diboson	30 513	7 629	38 142	0.93%
vjets	8 742	2 186	10 928	0.27%
Total	3 263 767	815 940	4 079 707	100%

The **higgsbkg** class merges ggH , VBF, $ggZH$, ZH , WH , $t\bar{t}H$ and $H + b$; **diboson** merges WW , WZ and ZZ ; **vjets** merges DY+jets and W +jets. The split is 80:20 per process, so the class proportions are the same in both sets. With 1 215 signal events in the test set, the statistical uncertainty on a test-set AUC near 0.97 is approximately 0.005; AUC differences smaller than this are not significant.

11.2 Input variables

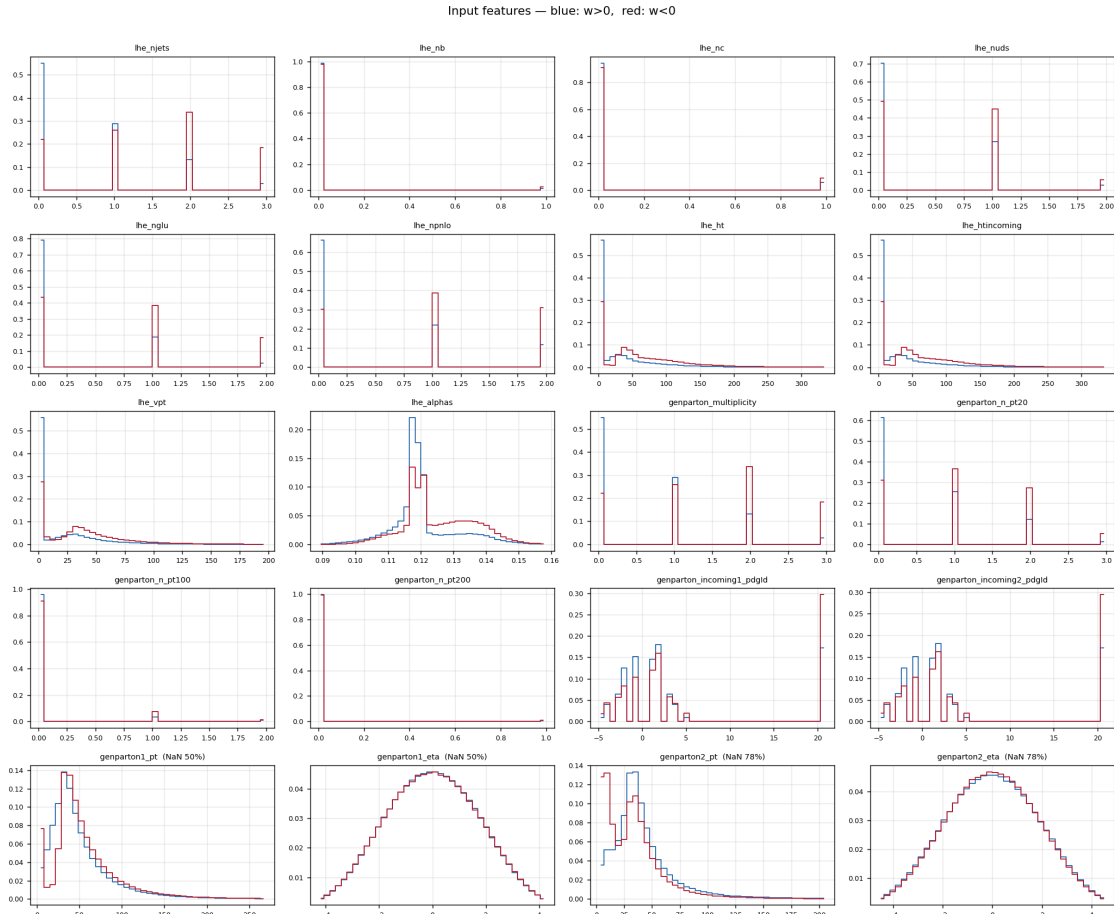


Figure 1: Distributions of the MVA input variables for signal and the principal backgrounds, after the base selection.

11.2 Input variables

The network takes 27 inputs: 15 kinematic variables, the 11 one-hot two-dimensional c-tagging category flags of Sec. 13, and the secondary-vertex multiplicity. The kinematic set follows the Run 2 analysis.

11.3 Use of the training events in the fit

Table 23: MVA input variables. The one-hot c-tagging flags replace the raw PNet CvL/CvB scores, so that the c-tagging scale factors of Sec. 13 act on the same categories the network sees.

Variable	Meaning
dilepton_pt	$p_T(\ell\ell)$
lepton1_pt	$p_T(\ell_1)$
lepton2_pt	$p_T(\ell_2)$
cjet_cand_pt	p_T of the c-jet candidate
met_pt	p_T^{miss}
mtl1	$m_T(\ell_1, p_T^{\text{miss}})$
mtl2	$m_T(\ell_2, p_T^{\text{miss}})$
dilepton_mass	$m(\ell\ell)$
delta_R_ll_l1	$\Delta R(\ell\ell, \ell_1)$
delta_R_ll_l2	$\Delta R(\ell\ell, \ell_2)$
delta_R_ll_c	$\Delta R(\ell\ell, c)$
delta_phi_l1PlusMET_c	$\Delta\phi(\ell_1 + p_T^{\text{miss}}, c)$
delta_phi_l1_MET	$\Delta\phi(\ell_1, p_T^{\text{miss}})$
delta_phi_l2_MET	$\Delta\phi(\ell_2, p_T^{\text{miss}})$
nSV	number of secondary vertices
cjet_cand_ctag2d_*	11 one-hot c-tag category flags

TODO: state whether any input sculpts the background shape. The discriminant is built from $m(\ell\ell)$ and m_T , which are correlated with the $H \rightarrow WW$ kinematics; a check that the background templates are not sculpted into a peak under the signal is needed.

11.3 Use of the training events in the fit

The fit categories are the `argmax` of the network output, and the templates entering the fit are built from all selected events, including the 80% used for training. Any residual overtraining therefore sculpts the template shapes directly, and no nuisance in the card covers it.

TODO: demonstrate that this is not an issue, either by scoring every event with a k -fold ensemble so that no event is scored by a network that saw it, or by overlaying the signal-region template shapes built from the training and test subsets and showing they agree within their statistical uncertainties. A train-versus-test ROC overlay should be shown alongside Fig. 2.

11.4 Model architecture

The classifier is a feed-forward multilayer perceptron, `SimpleMLP_MultiClass`, with batch normalisation and dropout between hidden layers.

Table 24: Model and training hyperparameters.

Parameter	Value
Architecture	$N_{\text{feat}} \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow N_{\text{class}}$
Regularisation	BatchNorm, Dropout $p = 0.2$
Optimiser	Adam, $\text{lr} = 10^{-3}$
Batch size	1024
Epochs	30 (v11), 50 (v32)
Class weighting	inverse frequency

11.5 Six-class training (v11)

The baseline training uses six classes,

$$\{\text{hplusc}, \text{higgsbkg}, t\bar{t}, \text{st}, \text{diboson}, \text{vjets}\},$$

where `higgsbkg` merges all non-signal Higgs production modes ($H + b$, ggH , VBF, ZH , $ggZH$, WH , $t\bar{t}H$) into a single class. Standard cross-entropy is used with inverse-frequency class weights.

An earlier attempt with thirteen flat classes (v10) failed, giving $\text{AUC}(\text{hplusc vs all}) \approx 0.51$: with extreme class imbalance across thirteen targets the network did not learn useful separation. Merging the Higgs modes into a single background class resolves this.

Table 25: Test-set AUC for the six-class v11 training.

Discriminant	AUC
hplusc vs all	0.966
hplusc vs $t\bar{t}$	0.976
hplusc vs diboson	0.931
hplusc vs higgsbkg	0.882

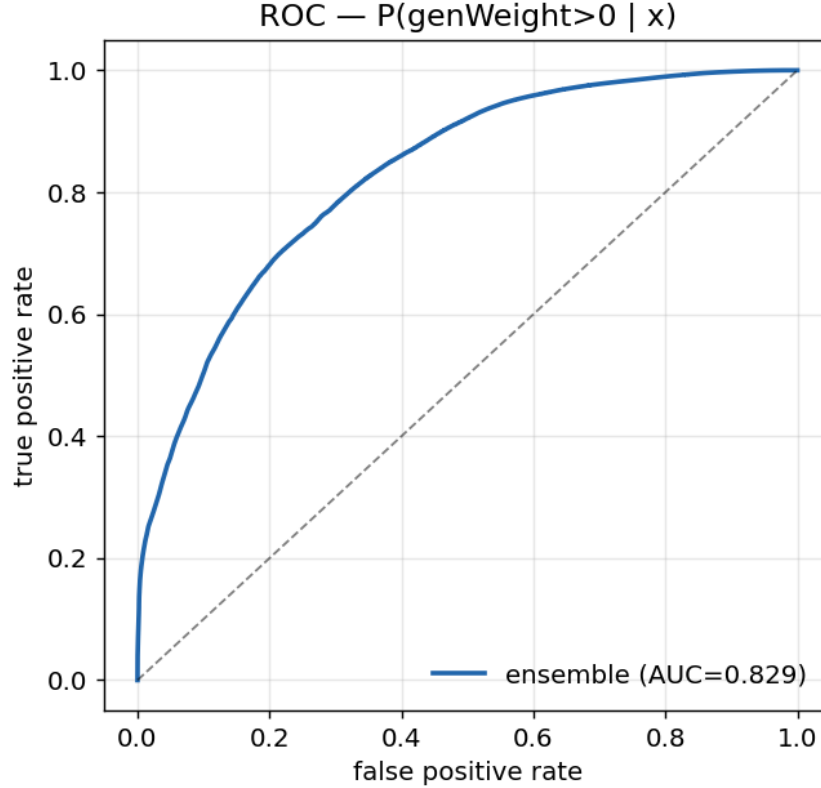


Figure 2: Test-set ROC curves for the six-class **v11** training. The **hplusc** vs all discriminant reaches an AUC of 0.966.

The lowest separation is against **higgsbkg**, which is expected: ggH produced in association with a heavy-flavour jet is kinematically similar to the signal, and this irreducible similarity sets the ceiling on the achievable sensitivity.

11.6 κ -HCE training (v32)

The six-class scheme discards the distinction between Higgs production modes. To recover that information without returning to the failure mode of **v10**, a hierarchical cross-entropy loss with a learned grouping is used over the full thirteen classes.

The loss has three terms,

$$\mathcal{L} = \mathcal{L}_{\text{group}} + K_{\text{fine}} \mathcal{L}_{\text{fine}} + \lambda \mathcal{L}_{\text{sig}},$$

where $\mathcal{L}_{\text{group}}$ learns the broad class structure, $\mathcal{L}_{\text{fine}}$ resolves the classes within the signal-like group, and $\mathcal{L}_{\text{sig}}$ optimises the D_κ discriminant directly.

The grouping is not fixed by hand. At each forward pass the cosine similarity $\cos \theta_i$ between the weight vector of class i and that of the signal class is computed; classes with $\cos \theta_i > 0$ form the signal-like group. The soft membership is

$$\alpha_i = \sigma(\cos \theta_i / \tau), \quad \tau = 0.3.$$

so that κ is fixed by a learned, physically interpretable quantity rather than being a free hyperparameter.

Table 26: Learned cosine similarity to the $H + c$ class and the corresponding soft group membership α at the end of v32 training.

Class	$\cos \theta$	α ($\tau = 0.3$)	Group
$H + c$	+1.000	0.966	signal
$H + b$	+0.355	0.765	+ cos
VBF	+0.277	0.716	+ cos
ggH	+0.097	0.580	+ cos
V+jets	-0.244	0.307	- cos

The grouping evolves during training: at the first epoch most classes have $\cos \theta > 0$, and by the final epoch only $H + b$, ggH and VBF remain in the signal-like group. This acts as a curriculum, with broad multi-class separation learned first and signal-like discrimination refined later. Diboson is approximately orthogonal to the signal, while $t\bar{t}$ and single top acquire strongly negative similarity, consistent with their role as the genuine backgrounds.

Table 27: Test-set AUC for the κ -HCE training, compared with v11. Both trainings use 30 epochs and identical optimiser settings. The statistical uncertainty on a test-set AUC near 0.97 is approximately 0.005.

Discriminant	v11	v32
hplusc vs all	0.966	0.975
higgs vs all	—	0.796
hplusc given higgs	—	0.887

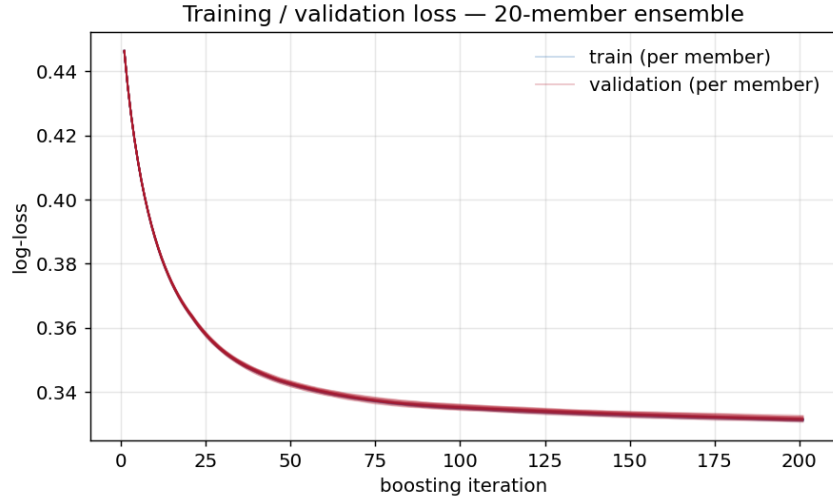


Figure 3: Training and validation loss for the κ -HCE training. The group, fine and significance terms are shown separately.

11.7 Choice of training for the fit

The v32 training gives the better raw signal discrimination, but the six-class v11 output is used to define the categories entering the fit (Sec. 16). The six classes map one-to-one onto the six processes of the datacard, so the `argmax` of the network output partitions the events into a signal region and five control regions with no additional cuts. The finer v32 scheme is retained for studies that require the Higgs sub-classes to be separated.

12 The κ -HCE Loss

This section derives the hierarchical cross-entropy loss with dynamic, learned groups (“ κ -HCE”) used for the thirteen-class training of Sec. 11. The loss replaces manually defined class hierarchies with groups derived from the geometry of the classifier’s own output layer, and couples them to a differentiable significance term.

12.1 Notation

Symbol	Meaning
C	number of classes (13)
s	signal class index ($H + c$)
$z \in \mathbb{R}^C$	network logits for one event
$p_i = \text{softmax}(z)_i$	posterior $P(\text{class } i \mid x)$
$W \in \mathbb{R}^{C \times h}$	weight matrix of the final linear layer, $h = 32$
W_j	row j of W : the learned template vector for class j
y	truth label
N_j	number of training events in class j
τ	temperature of the soft group membership
λ, k_{fine}	weights of the significance and fine terms

The logit for class j is $z_j = W_j \cdot \phi(x) + b_j$, with $\phi(x)$ the penultimate activation. Each row W_j is therefore the direction in feature space that the network associates with class j , and two classes whose templates point in similar directions are confusable. This geometric observation is what the loss exploits.

12.2 Dynamic groups from cosine similarity

The cosine similarity between the signal template and every class template is

$$c_j = \cos(W_s, W_j) = \frac{W_s \cdot W_j}{\|W_s\| \|W_j\|}, \quad c_s = 1, \quad (12.1)$$

recomputed at every forward pass from the live weights, so that the grouping evolves during training. The hard split used by the group term is

$$\mathcal{P} = \{j : c_j \geq 0\}, \quad \mathcal{N} = \{j : c_j < 0\}, \quad n_{\mathcal{N}} = |\mathcal{N}|, \quad (12.2)$$

with $\mathcal{P}$ the signal-like group. The soft membership used by the fine and significance terms is

$$\alpha_j = \sigma\left(\frac{c_j}{\tau}\right) = \frac{1}{1 + e^{-c_j/\tau}} \in (0, 1), \quad (12.3)$$

where α_j plays the role of κ : the degree to which class j is treated as signal-like. For $\tau = 0.3$, $\alpha > 0.5$ once $c_j > 0$ and $\alpha < 0.05$ by $c_j < -0.6$.

α is computed from a detached copy of c_j , so it acts as a read-out gate rather than a free parameter driven by the loss. The template geometry is shaped only by the cross-entropy terms, and the significance term cannot move α . This is what avoids the collapse ($\kappa \rightarrow 0$) and saturation ($\kappa \rightarrow 1$) pathologies of earlier learned- κ losses.

12.3 The three terms

$$\mathcal{L} = \mathcal{L}_{\text{group}} + k_{\text{fine}} \mathcal{L}_{\text{fine}} + \lambda \mathcal{L}_{\text{sig}}. \quad (12.4)$$

12.3.1 Group term

The $\mathcal{P}$ classes are collapsed into a single merged column while each $\mathcal{N}$ class is kept separate, giving grouped probabilities $q \in \mathbb{R}^{n_{\mathcal{N}}+1}$:

$$q_k = p_{\mathcal{N}_k} \ (k < n_{\mathcal{N}}), \quad q_{n_{\mathcal{N}}} = \sum_{j \in \mathcal{P}} p_j. \quad (12.5)$$

With $g(y)$ mapping an event's truth label to its group index,

$$\mathcal{L}_{\text{group}} = - \frac{\sum_x w_{g(y)} \log q_{g(y)}(x)}{\sum_x w_{g(y)}}, \quad (12.6)$$

a weighted negative log-likelihood with inverse-frequency group weights

$$w_k = \widehat{\text{invfreq}}(\mathcal{N}_k) \ (k < n_{\mathcal{N}}), \quad w_{n_{\mathcal{N}}} = \frac{1}{\sum_{j \in \mathcal{P}} N_j}, \quad \widehat{\text{invfreq}}(j) = \frac{\sum_i N_i}{N_j} \quad (12.7)$$

normalised so that $\max_j = 1$.

The positive classes are merged rather than given individual columns because a merged column contributes no gradient that pushes the $\mathcal{P}$ templates apart. Individual columns would repel their weight vectors and destabilise the very cosine structure that defines the groups; merging keeps the signal-like cluster cohesive.

12.3.2 Fine term

An α -gated conditional distribution is built over the classes,

$$\tilde{p}_i(x) = \frac{\alpha_i p_i(x)}{\sum_j \alpha_j p_j(x)}, \quad (12.8)$$

and the per-event fine loss, weighted by the α of the event's own true class, is

$$\ell_{\text{fine}}(x) = -\widehat{\text{invfreq}}(y) \log \tilde{p}_y(x), \quad \mathcal{L}_{\text{fine}} = \frac{\sum_x \alpha_{y(x)} \ell_{\text{fine}}(x)}{\sum_x \alpha_{y(x)}}. \quad (12.9)$$

α appears twice: inside $\tilde{p}$, which restricts the denominator to the signal-like region, and as the per-event weight $\alpha_{y(x)}$, which down-weights background events. Together these make the term focus on separating the signal from its nearest neighbours.

12.3.3 Significance term

The κ discriminant is defined with α in the role of κ ,

$$D(x) = \frac{p_s(x)}{p_s(x) + \sum_{j \neq s} \alpha_j p_j(x)} \in (0, 1]. \quad (12.10)$$

Signal-like classes ($\alpha_j \rightarrow 1$) remain in the denominator and dilute D , so that D grows only when p_s genuinely dominates its near neighbours; background-like classes ($\alpha_j \rightarrow 0$) drop out and cannot inflate it. The term is a batch mean on each half,

$$\mathcal{L}_{\text{sig}} = \frac{1}{n_{\text{sig}}} \sum_{x: y=s} -\log D(x) + \frac{1}{n_{\text{bkg}}} \sum_{x: y \neq s} \log(1 + D(x)). \quad (12.11)$$

The first term diverges as $D \rightarrow 0$, giving a strong pull to raise p_s on true signal. The second is bounded below by zero and suppresses p_s on backgrounds without dominating; this is the reason for using $\log(1 + D)$ rather than a $\log B$ term, which is unbounded from below. Normalising each half by its own count removes the dependence on batch composition that caused earlier summed significance losses to spike on signal-heavy batches.

12.4 Implementation

The loss is implemented in `utils/loss/HierarchicalCrossEntropyLoss.py` (`_kappa_hce_forward`, mode "`kappa_hce`") within the `b-hive` framework, configured by `config/HPlusCHToWW_kappa_hce.yml`. The final training uses $\lambda = 1$, $\tau = 0.3$, $k_{\text{fine}} = 7$, batch size 1024, learning rate 10^{-3} and 30 epochs. The six-class `v11` training uses the same optimiser settings and the same number of epochs, so the AUC comparison of Table 27 is not confounded by training length.

12.5 Status of this loss in the analysis

The κ -HCE training is not used to define the fit categories; the six-class `v11` output is, for the reasons given in Sec. 11. This section documents the method and its validation, and the thirteen-class output is retained for studies that require the Higgs sub-classes to be separated.

TODO: either quote a limit obtained with the κ -HCE categories, so that the improved AUC can be related to sensitivity, or move this section to an appendix. An ablation is also needed: the reported gain ($0.966 \rightarrow 0.975$) is a single number for a three-term loss with three hyperparameters, and the contribution of the group, fine and significance terms has not been separated. Note that the AUC difference is comparable to the test-set statistical uncertainty of about 0.005 quoted in Sec. 11.

13 Charm Tagging

13.1 The two-dimensional scheme

ParticleNet provides two charm discriminants per jet: CvL , separating charm from light flavour, and CvB , separating charm from bottom. Rather than applying a single working-point cut, the calibration partitions the whole (CvL, CvB) plane into eleven categories,

$$L0, \quad C0-C4, \quad B0-B4,$$

including the untagged $L0$ region. The category index is computed from the CvL and CvB values already stored per jet in NanoAOD, so no reprocessing of the input samples is required. The bin edges are those frozen in the official calibration and must not be altered between runs, since changing them invalidates the scale-factor lookup.

The scheme is defined on the two axes CvL and CvB .

TODO: the claim that these two axes are sufficient and that no third discriminant is needed requires a supporting reference or study. Cite the BTV recommendation the scheme is taken from, or remove the claim.

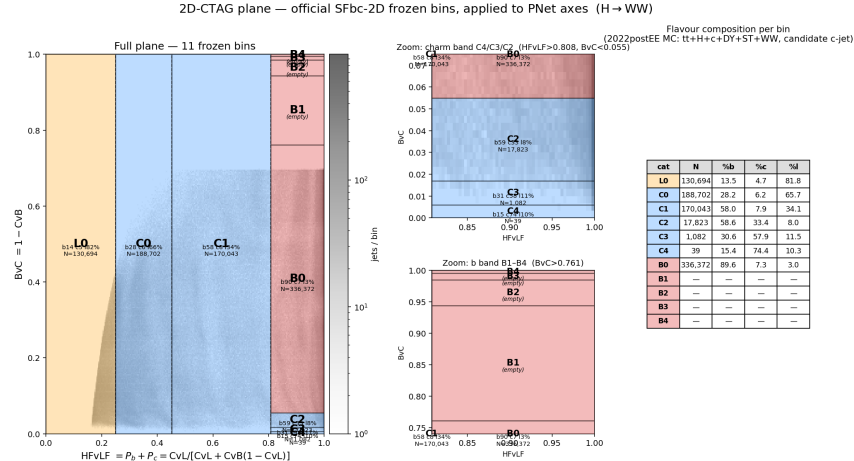


Figure 4: The eleven two-dimensional c -tagging categories in the CvL – CvB plane.

13.2 Occupancy in this analysis

The category refers to one specific jet, the *candidate c-jet*, defined as the jet with the highest CvL score among those passing the jet selection. That selection already requires the medium c -tagging working point, so the candidate is by construction c -tagged.

Only seven of the eleven categories are populated after the $e\mu$ selection. The b -rich $B1$ – $B4$ categories are empty, which is a consequence of this definition rather than a statement about the event content: $t\bar{t}$ events do contain b jets in abundance, but a jet failing the medium c -tag working point is never chosen as the candidate, so it cannot

enter a B category. The scheme itself is designed for a broader phase space in which no working-point requirement is imposed upstream.

Table 28: Flavour composition of the five most populated categories.

Category	Jets	c [%]	b [%]	light [%]
L0	130 694	13.5	4.7	81.8
C0	188 702	28.2	6.2	65.7
C1	170 043	58.0	7.9	34.1
C2	17 823	58.6	33.4	8.0
C3	1 082	30.6	57.9	11.5

13.3 Application of the scale factors

The per-category scale factors are applied natively in the processor through the `Ctag2DCorrector` class; the earlier post-hoc application script is retired. Because the scheme spans the full plane including the untagged L0 category, the correction is defined for every jet and the machinery remains valid even for a selection with no working-point requirement.

The per-category central values and their uncertainty band are propagated as a single up/down variation of the event weight, entering the datacard as the shape nuisance `CMS_ctag2d_2022`. This is presently one nuisance covering the entire plane; a decomposition into independent components remains to be done.

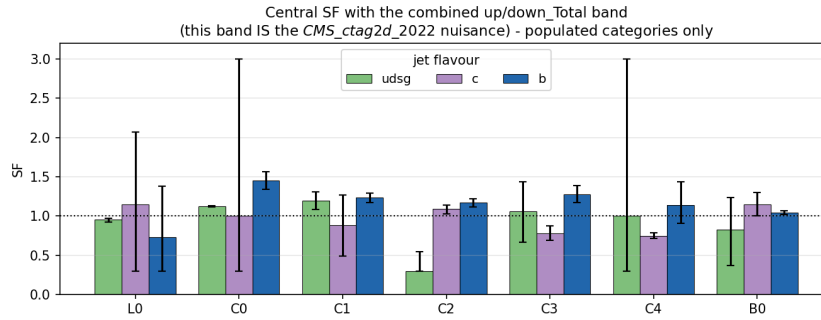


Figure 5: Scale-factor uncertainty band per category.

13.4 Effect on the limit

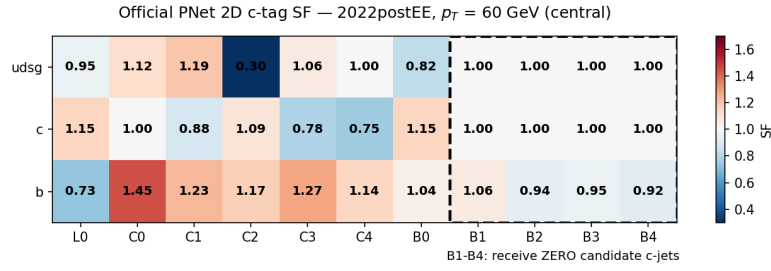


Figure 6: Two-dimensional c -tagging scale factors across the eleven categories.

13.4 Effect on the limit

The comparison with and without the scale factor was performed as a controlled test: the no-SF variant was built from the pre-SF backup templates through the same datcard builder on the same day, with only the correction toggled.

Table 29: Effect of the two-dimensional charm-tagging scale factor.

Configuration	Full	Stat. only	Frozen autoMCStats
No scale factor	1150	668	905
With CMS_ctag2d_2022	1164	676	930

Applying the calibration degrades the expected limit by 14. Part of this is the additional nuisance parameter, but that is not the whole explanation: a scale factor also moves the central prediction, and the stat-only limit changes from 668 to 676 when it is applied, which an extra-nuisance-only effect could not do. The purpose of applying the calibration is correctness rather than sensitivity.

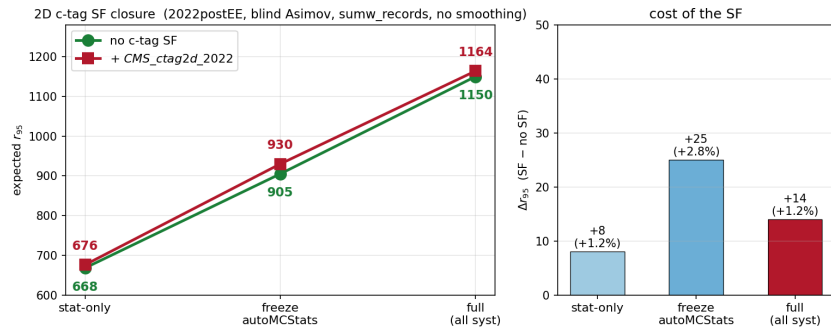


Figure 7: Closure of the two-dimensional scale factors: ratio of corrected simulation to data.

13.5 Use in the MVA

The eleven categories are also supplied to the MVA as one-hot encoded inputs, so that the network has access to the full two-dimensional plane rather than a binary tagged/untagged decision. This brings the number of input features to 26.

14 Negative-Weight Reweighting for V+jets

14.1 Origin and consequence of negative weights

An NLO cross section is computed as a sum of real, virtual and Born contributions,

$$\sigma_{\text{NLO}} = \int d\Phi_{n+1} R + \int d\Phi_n V + \int d\Phi_n B, \quad (14.1)$$

in which R and V are separately divergent and cancel only in the sum. Numerical integration requires subtraction terms S ,

$$\sigma_{\text{NLO}} = \int d\Phi_{n+1} (R - S) + \int d\Phi_n \left(V + \int_1 S \right) + \int d\Phi_n B, \quad (14.2)$$

where each bracket is finite but not positive definite. Events sampled from them therefore carry weights of either sign. Negative weights are an unavoidable feature of a matched NLO calculation rather than a generator defect.

The consequence for an analysis is a loss of effective statistics. For a bin B the yield estimator

$$\hat{N}_B = \sum_{i \in B} w_i, \quad w_i = \pm |w_i|, \quad (14.3)$$

is unbiased, but its variance scales with $\sum_i w_i^2 = \sum_i |w_i|^2$, that is with the *unsigned* statistics, while the mean is the much smaller signed sum. The effective sample size

$$n_{\text{eff}} = \frac{\left(\sum_i w_i \right)^2}{\sum_i w_i^2} \quad (14.4)$$

is then far below the raw event count. In the V+jets samples used here 16.4% of events carry a negative weight, and in sparsely populated bins the cancellation can be nearly complete: one DY bin was measured at 0 ± 41 events from $\pm 79\,000$ weights cancelling, producing a bin with no content and a large finite uncertainty.

14.2 The reweighting identity

Any signed density may be written as the difference of two positive, normalised densities,

$$p(\vec{x}) = a p_+(\vec{x}) - b p_-(\vec{x}), \quad a, b \geq 0, \quad (14.5)$$

where p_+ and p_- are the densities of positive- and negative-weight events and normalisation of p requires $a - b = 1$. The probability that an event at $\vec{x}$ carries a positive weight is

$$P_+(\vec{x}) = \frac{a p_+(\vec{x})}{a p_+(\vec{x}) + b p_-(\vec{x})}. \quad (14.6)$$

Writing $u(\vec{x}) \equiv a p_+(\vec{x}) + b p_-(\vec{x})$ for the unsigned density gives $ap_+ = P_+u$ and $bp_- = (1 - P_+)u$, and substituting into the decomposition,

$$p(\vec{x}) = P_+u - (1 - P_+)u = (2P_+(\vec{x}) - 1) u(\vec{x}), \quad (14.7)$$

so that

$$\boxed{p(\vec{x}) = g(\vec{x}) u(\vec{x}), \quad g(\vec{x}) \equiv 2P_+(\vec{x}) - 1} \quad (14.8)$$

Since u is precisely the density sampled by $\sum_i |w_i|$, for any observable f

$$\int f(\vec{x}) p(\vec{x}) d\vec{x} = \int f(\vec{x}) g(\vec{x}) u(\vec{x}) d\vec{x}, \quad (14.9)$$

and therefore

$$\sum_i w_i f(\vec{x}_i) \hat{=} \sum_i |w_i| g(\vec{x}_i) f(\vec{x}_i). \quad (14.10)$$

The two estimators describe the same distribution. The reweighting is thus $w_i \rightarrow |w_i| g(\vec{x}_i)$: an algebraic identity, not an approximation, for the true P_+ .

14.3 Variance reduction

Because $|g| \leq 1$, each term is reduced in magnitude and, more importantly, no cancellation occurs:

$$V\left[\sum_i |w_i| g_i\right] = \sum_i |w_i|^2 g_i^2 \leq \sum_i w_i^2, \quad (14.11)$$

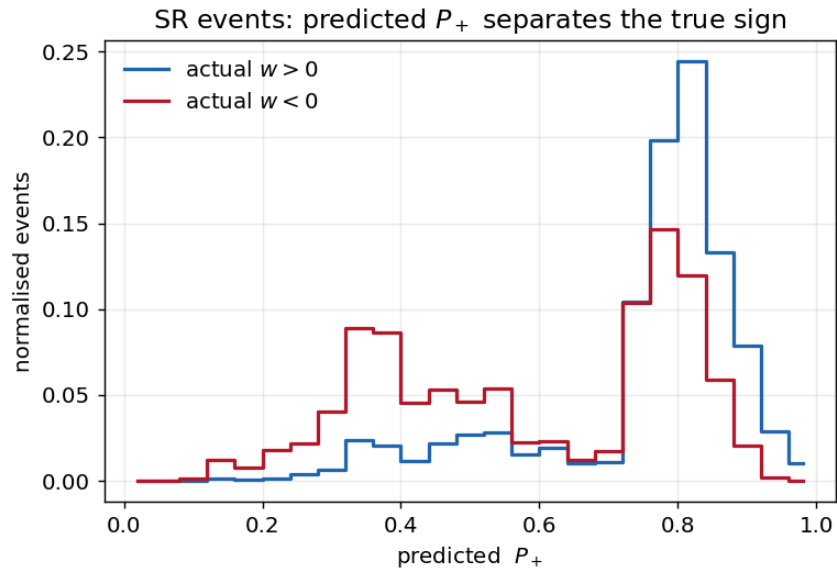
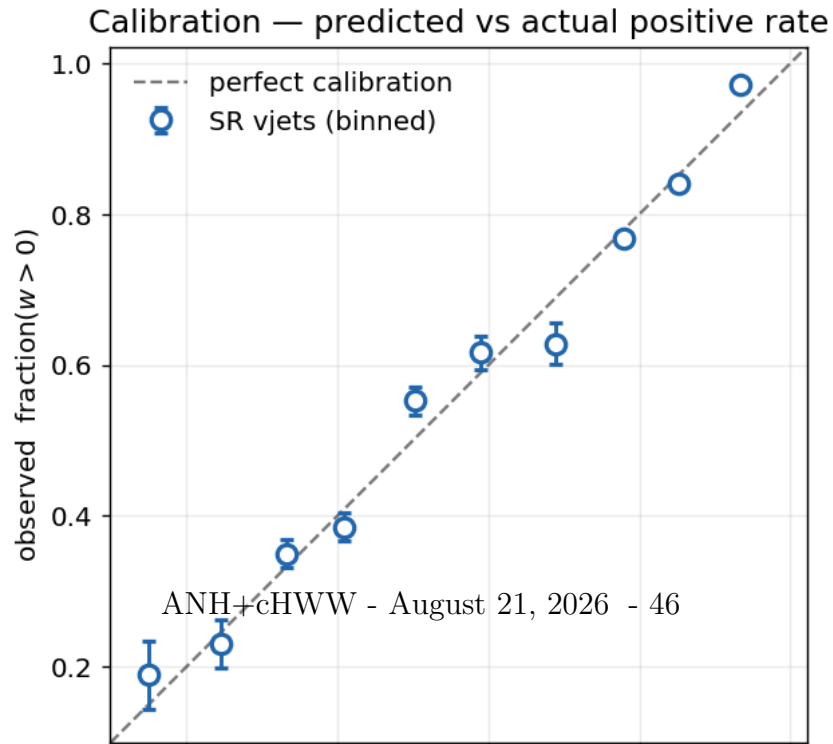
with equality only where $|g| = 1$, that is where the weight sign is deterministic. Measured on the samples used here, $\langle g \rangle = 0.672 = 2(0.836) - 1$, consistent with the global positive fraction, with g spanning $[-0.991, 0.993]$ and therefore remaining inside the physical range.

14.4 Estimation of P_+

P_+ is not known analytically and is estimated with a classifier, $\hat{g} = 2\hat{P}_+ - 1$. This is the only approximation in the method: closure ceases to be an algebraic guarantee and becomes a property that must be measured.

Table 30: Classifier used to estimate P_+ .

Setting	Value
Model	HistGradientBoostingClassifier, 20-model ensemble
Target	$\text{sign}(w)$
Features	20, all generator level
Training events	9.8×10^6
Ensemble AUC	0.829

Figure 8: Distribution of P_+ separately for events with genuinely positive and negative generator weights.

The inputs are exclusively generator-level quantities: `lhe_njets`, `lhe_nb/nc/nuds/nglu`, `lhe_ht`, `lhe_vpt`, `lhe_alphas`, `genparton_multiplicity`, `genparton_n_pt20/100/200`, the incoming parton PDG identifiers, and the transverse momenta and pseudorapidities of the two leading partons. Reconstruction-level variables are excluded by construction: the identity $p = g u$ holds in the variables the generator actually sampled, and u would no longer be the density sampled by $\sum |w|$ if $\vec{x}$ contained reconstructed quantities.

An AUC of 0.829 is obtained on an intrinsically stochastic target, since the sign of a given event's weight is not a deterministic function of its kinematics and a perfect classifier cannot exist even in principle.

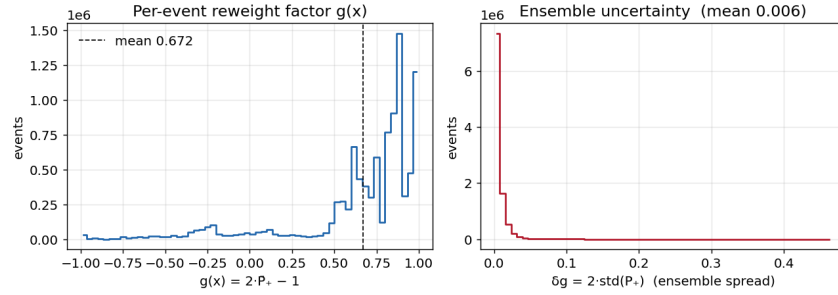


Figure 10: The reweighting factor $g = 2P_+ - 1$ and its ensemble uncertainty δg .

14.5 Training region

The signal region contains too few V +jets events to train on, that starvation being the problem the method addresses. The classifier is therefore trained on a much larger region defined by the base selection with a veto on the $e\mu$ signal-region topology, and applied to the signal region.

The veto makes the two sets disjoint by construction: an event that does not contain exactly one $e\mu$ pair can never enter the signal region, so no event-level bookkeeping is required and no overfitting bias arises. The regions are disjoint in *events* but not orthogonal in phase space; g must be interpolated across the signal region's support rather than extrapolated, so a training region chosen to be kinematically opposite to the signal region would be actively harmful.

14.6 Closure and renormalisation

Closure in the training region is 0.994 over 9.8×10^6 events. In the signal region, which is a small and kinematically biased corner of the training domain, the per-event g does not average exactly to the local positive fraction:

Table 31: Signal-region closure before renormalisation, per process.

Process	$\sum w$	$\sum w g$	$\sum w g / \sum w$
DY (both mass bins)	1.014×10^8	1.027×10^8	1.013
$W \rightarrow \ell \nu$	5.238×10^7	5.876×10^7	1.122

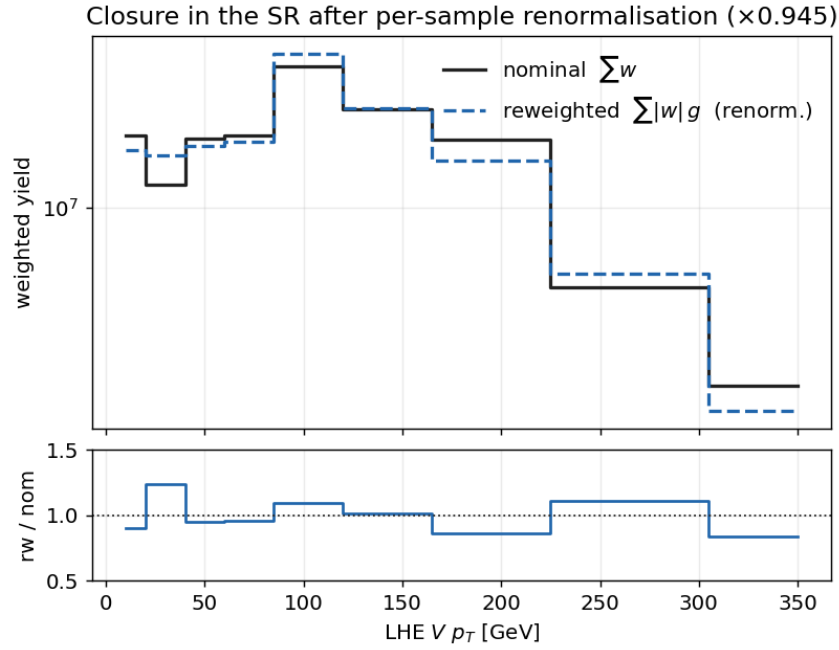
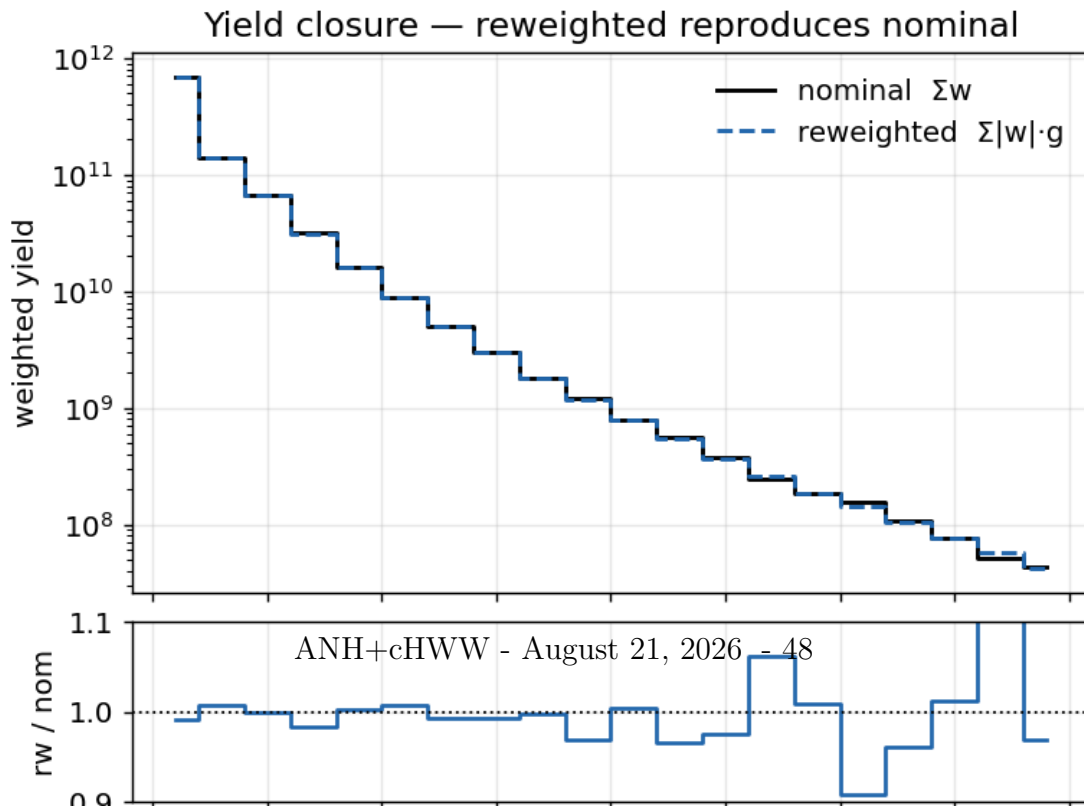


Figure 11: Closure after per-dataset renormalisation, which removes the residual normalisation offset.



The reweighted template is therefore renormalised to the nominal yield, separately for each dataset, in the datacard builder. The yield is restored exactly, while the variance reduction, which arises from the per-bin spread rather than the integral, is unaffected.

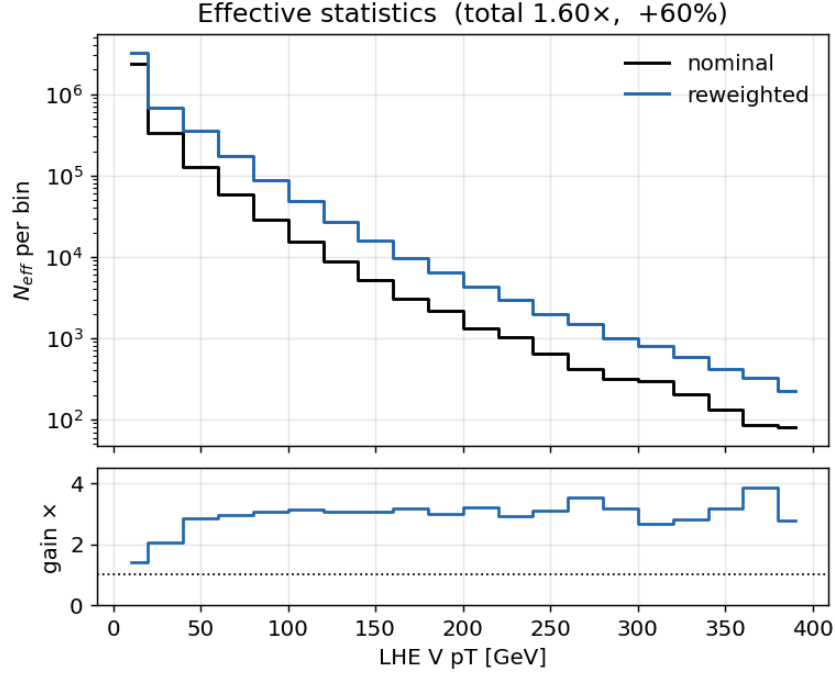


Figure 13: Gain in effective statistics n_{eff} from the reweighting.

14.7 Uncertainty

The spread of the twenty-model ensemble is propagated as a shape nuisance, `CMS_negrw_vjets`, with mean $\delta g = 0.006$. Two per-event columns are written alongside the nominal weight:

$$\text{weight_negrw} = 2 \bar{P}_+(\vec{x}) - 1 = g, \quad \text{weight_negrw_std} = 2 \text{std}[P_+(\vec{x})] = \delta g. \quad (14.12)$$

The reweighting is applied only to the V+jets samples for which the ensemble was trained, using an anchored dataset match; a substring match would also select the WH signal sample and silently reweight a Higgs template with a V+jets model. In the fit `CMS_negrw_vjets` profiles to a negligible impact.

14.8 Applicability to the jet-binned samples

The classifier described here was trained on the inclusive W +jets sample, whose negative weight fraction is 16.4%. The jet-binned samples that replace it (Sec. 15) have a

substantially larger negative fraction, 10.2%, 25.7% and 34.7% for the $0J$, $1J$ and $2J$ bins respectively, or about 20.9% weighted by event count.

TODO: confirm whether the ensemble was retrained on the jet-binned samples. If it was not, the classifier is being evaluated outside the sample it was trained on, and the $2J$ bin — which supplies 69% of the surviving signal-region events and has both the worst negative fraction and the worst n_{eff}/N — is the case where that matters most. Either retrain per jet bin or demonstrate that the reweighting closes on each bin separately.

14.9 Limitations of the uncertainty estimate

`CMS_negrw_vjets` is the spread of an ensemble of twenty classifiers differing only in their random seed, trained on the same events with the same features. It therefore measures the seed variance of the fit, and not the dominant error, which is the systematic mismodelling of P_+ by a classifier built from twenty aggregate generator-level quantities. The closure of Table 31 is direct evidence that the true uncertainty is larger than δg : the $W \rightarrow \ell\nu$ signal-region closure is 1.122, a 12% effect, against an ensemble spread of $\delta g \approx 0.006$.

The reweighting exchanges a statistical uncertainty for a systematic one, and the systematic is presently assigned as negligible. That trade is not yet justified.

TODO: replace the ensemble spread with an uncertainty that covers the observed non-closure, for example by taking the per-sample closure deviation as an additional nuisance, or by a k -fold estimate that varies the training set rather than the seed. Also show the reweighted and renormalised signal-region *shape* against the nominal signed-weight shape: the renormalisation restores the integral, but the fit uses the shape.

14.10 Relation to existing work

The algebraic identity underlying this reweighting is the one set out in Ref. [2], which the implementation described here follows. The same technique has been developed independently within CMS as PRISM-PCA [4], applied to the VHbb/cc analysis, using a Particle Transformer model with particle-level inputs and a PCA-based treatment of the reweighting uncertainty. The implementation described here is an independent application using a gradient-boosted decision tree on aggregate generator-level features, and reproduces a comparable gain in effective statistics.

15 W+jets Sample Choice

15.1 Motivation

V+jets is the process for which the effective Monte Carlo statistics are weakest in the signal region. Before the change described here, V+jets contributed $n_{\text{eff}} = 280$ in `SR_hplusc`, corresponding to a relative statistical uncertainty of 5.98%, against $\leq 1.1\%$ for every other background. The `autoMCStats` nuisances associated with this were the dominant systematic effect in the fit.

AN-23-102 [3] does not use the inclusive NLO W+jets sample for the same reason, describing it as having a large negative-weight fraction, and instead stitches jet-binned and p_T -binned samples.

15.2 Sample replacement

The inclusive `WtoLNU-2Jets` sample is replaced by the three jet-binned NLO samples. Cross sections are taken from XSDB.

Table 32: Inclusive and jet-binned W+jets samples for 2022postEE. The negative-weight fractions were verified directly on the NanoAOD and agree with the database values to about 1%.

Sample	σ [pb]	Events	Files	Neg. weights
<code>WtoLNU-2Jets_0J</code>	55 760	678.4×10^6	3432	10.2%
<code>WtoLNU-2Jets_1J</code>	9 529	522.6×10^6	2669	25.7%
<code>WtoLNU-2Jets_2J</code>	3 532	344.6×10^6	2135	34.7%
Sum	68 821	1545.5×10^6	8236	
<i>Inclusive (replaced)</i>	<i>67 710</i>	<i>281.5×10^6</i>	<i>381</i>	<i>16.1%</i>

The sum of the jet-binned cross sections exceeds the inclusive value by 1.6%, consistent with the expected spread from NLO merging and not indicative of double counting. Every event in the 0J sample was verified to have LHE $V_{p_T} < 100$ GeV, so the three jet-binned samples are self-contained and do not overlap with the p_T -binned samples, which are not included here. This means the region $V_{p_T} > 100$ GeV is not covered by the additional statistics.

TODO: quantify this gap — give the fraction of the signal-region V+jets yield with $V_{p_T} > 100$ GeV and its n_{eff} , so the size of the uncovered region is explicit. Explain also why the p_T -binned samples were not stitched in, given that AN-23-102 does exactly this.

15.3 Effective statistics

The raw event count increases by a factor 5.5, but the relevant quantity is the effective statistics per unit cross section:

Table 33: Effective luminosity of the inclusive and jet-binned samples. The three jet bins are exclusive, so their equivalent luminosities do not add; the combined figure is $\sum_i n_{\text{eff},i} / \sum_i \sigma_i = 8.5 \text{ fb}^{-1}$, not the sum 29.8 fb^{-1} of the individual entries.

Sample	n_{eff}/N	Equivalent lumi. [fb^{-1}]
0J	0.635	7.7
1J	0.237	13.0
2J	0.094	9.1
Combined		8.5
<i>Inclusive</i>	<i>0.460</i>	<i>1.9</i>

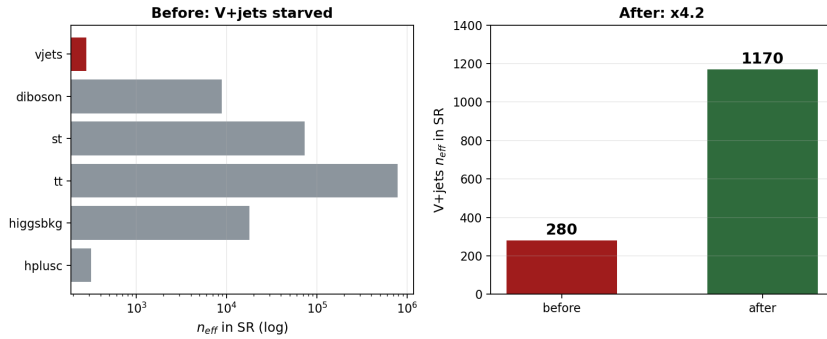


Figure 14: Effective statistics of the V +jets template before and after replacing the inclusive W +jets sample with the jet-binned samples.

The combined equivalent luminosity of 8.5 fb^{-1} against 1.9 fb^{-1} for the inclusive sample predicts a gain in effective statistics of $\times 4.5$, consistent with the $\times 4.2$ realised in the signal region (Table 34).

The inclusive sample devotes most of its cross section to zero-jet events, which almost never survive the requirement of at least one charm-tagged jet. Of the surviving signal region events, the 2J sample contributes 69% despite being the smallest of the three: the jet-binned samples are enriched in precisely the topologies the selection requires.

15.4 Effect on the fit

Table 34: Effect of the W +jets sample replacement.

Quantity	Inclusive	Jet-binned
Expected upper limit	1160	1034
V+jets n_{eff} (SR)	280	1170
V+jets rel. stat. uncertainty (SR)	5.98%	2.92%
V+jets yield (SR)	1507.7	1523.2

The yield changes by 1.0% while the statistical uncertainty is halved, which is the expected signature of a statistics improvement rather than a change in normalisation. The gain in n_{eff} is confined to V+jets: all other processes change by less than 13% and in both directions, consistent with the templates having been rebuilt.

16 Statistical Model

16.1 Categories

The six-class MVA of Sec. 11 defines the fit categories directly: the `argmax` of the network output assigns each event to exactly one region, and the value of the winning score is the discriminant binned within it. No additional selection is applied, so the regions are orthogonal by construction and every selected event is used.

Table 35: Composition of the six *argmax* categories.

Region	Events	Dominant process
SR_hplusc	21 552	$t\bar{t}$ 82.3%
CR_higgsbkg	9 220	$t\bar{t}$ 86.4%
CR_tt	44 199	$t\bar{t}$ 94.1%
CR_st	20 135	$t\bar{t}$ 87.6%
CR_diboson	6 585	$t\bar{t}$ 72.4%
CR_vjets	9 214	V+jets 43.2%

16.1 Categories

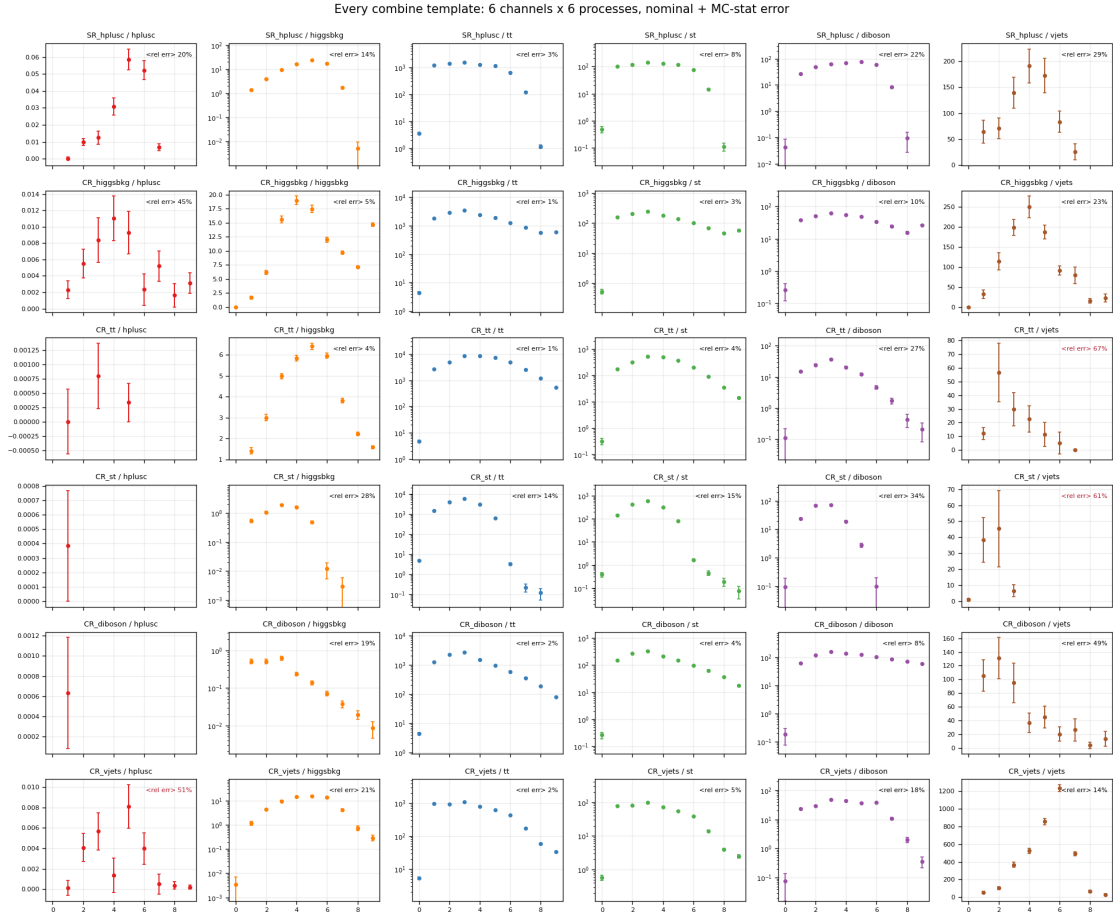


Figure 15: Prefit templates for each process in each channel.

16.2 The fit discriminant

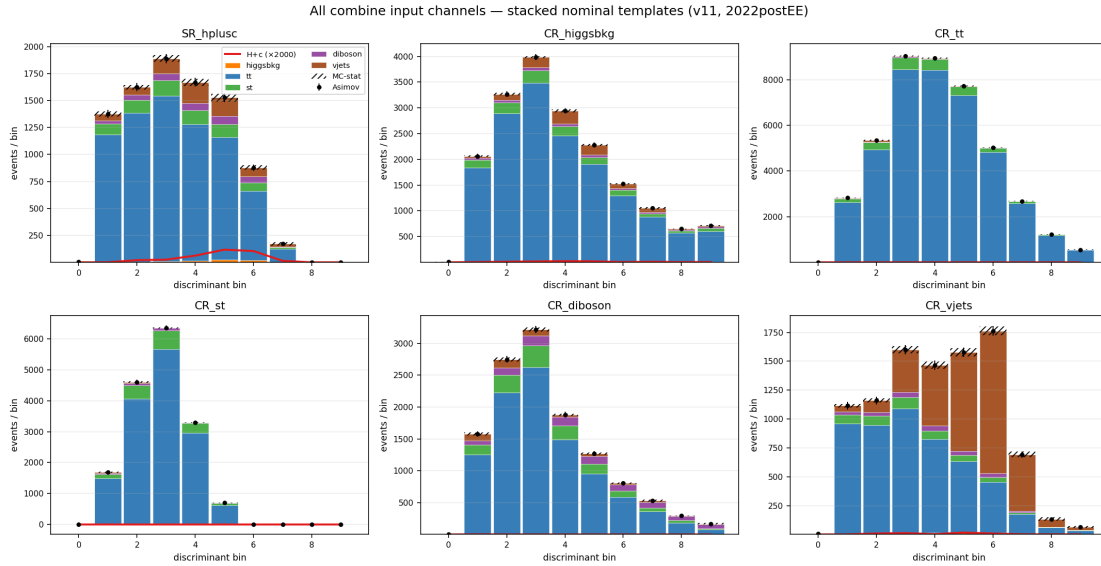


Figure 16: Stacked prefit templates for all six fit channels.

Four of the five control regions are dominated by $t\bar{t}$: the signal region is itself 82% $t\bar{t}$, since $t\bar{t}$ is the dominant process everywhere in an $e\mu$ final state. **CR_tt**, at 94.1% purity over 44 000 events, constrains the free-floating $t\bar{t}$ normalisation that covers 82% of the signal region. **CR_vjets** contains 59.6% of all V+jets events entering the fit.

Collapsing the control regions to single bins was tested and found to cost 33 units for three regions and approximately 50 for all five, so all six channels retain ten bins.

TODO: a control region is useful only if it constrains something. Show the prefit and postfit uncertainty on each background normalisation and demonstrate that **CR_st** and **CR_diboson** actually constrain single top and diboson respectively, rather than acting as four further $t\bar{t}$ regions with different score shapes. If they do not constrain their nominal process, they should be renamed or merged.

16.2 The fit discriminant

The quantity fitted in each category is the winning **argmax** score of the network. It is therefore bounded below by $1/C = 1/6$ by construction, and its distribution is shaped by the **argmax** operation itself rather than being a free likelihood ratio.

TODO: state the score range and the binning used in each category, whether the bins are equal-width or quantile-defined, and whether any signal bins are empty. The use of a winning-**argmax** score as the fit observable is unusual enough to warrant an explicit justification.

16.3 Comparison with data

TODO: no data/simulation comparison is shown anywhere in this note. The signal region is blind, but the five control regions are not, and at least **CR_tt** (94.1% $t\bar{t}$) and **CR_vjets** should be shown with data overlaid on the fitted discriminant. Without this there is no direct evidence that the simulation describes the selected phase space, which the MVA categories, the c-tagging calibration and the systematic model all rely on.

16.4 Datacard structure

Table 36: Datacard configuration.

Quantity	Value
Channels	6 (<code>imax</code> 6)
Processes	6 (<code>jmax</code> 5)
Bins per channel	10
Histograms	$1626 = 6 \times 6 \times (1 + 2 \times 22) + 6$
Observation	−1 in every channel (Asimov, blind)

Two entries require comment. The $t\bar{t}$ normalisation is a single free-floating parameter shared across all six channels,

```
rate_tt rateParam * tt 1.0 [0,5]
```

so that **CR_tt** determines the $t\bar{t}$ rate in the signal region. $t\bar{t}$ consequently carries no cross-section `lnN`, its normalisation being taken from data. Bin-by-bin Monte Carlo statistical uncertainties use the Barlow–Beeston-lite treatment,

```
<channel> autoMCStats 10
```

per channel. With a threshold of 10, bins with $n_{\text{eff}} > 10$ receive a single Gaussian nuisance while sparser bins receive individual Poisson terms.

16.5 Normalisation

Templates are produced from the analysis `parquet` files with

$$\text{read_scale} = \frac{\mathcal{L} \times \sigma}{\sum w}, \quad (16.1)$$

where the per-event generator weight enters after the selection through the weights container, while the denominator $\sum w$ is accumulated *before* any selection and written to per-chunk `sumw_records`. This distinction is essential: chunks that select zero events still contribute to the denominator, and reading $\sum w$ from the output file metadata instead undercounts it.

16.6 Object-shift systematics

Jet and lepton energy scale and resolution variations change the MVA score itself and therefore cannot be applied as event weights. Each is produced as a separate template tree, re-selected and re-scored through the network, giving twelve shifted directories. Inference coverage over every shifted directory is verified after each rebuild, since a partially scored set would leave the object-shift templates frozen at their nominal values.

17 Systematic Uncertainties

The card contains 22 shape nuisances, 9 rate (`lnN`) nuisances, one free-floating rate parameter and the bin-by-bin `autoMCStats` terms.

TODO: no trigger efficiency uncertainty appears in this list. If it is covered elsewhere in this note, cross-reference it here; if it is not, it is missing and should be added.

17.1 Shape uncertainties

Table 37: Weight-based shape nuisances.

Nuisance	Processes	Source
<code>pileup</code>	all MC	LUM POG pileup profile
<code>ps_isr, ps_fsr</code>	all except $t\bar{t}$	LHE parton-shower weights
<code>scalevar_muR, _muF, _muR_muF</code>	all except $t\bar{t}$	<code>LHEScaleWeight</code> , $\mu_{R,F} \times \{0.5, 2\}$
<code>lhe_pdf, lhe_alphaS</code>	all MC	NNPDF replicas, MC2Hessian
<code>muon_id, muon_iso</code>	all MC	MUO POG
<code>electron_id</code>	all MC	EGM POG
<code>electron_reco</code> ($\times 3$)	all MC	EGM POG, p_T -binned
<code>CMS_ctag2d_2022</code>	all MC	BTV two-dimensional SF
<code>CMS_ngrw_vjets</code>	V+jets	ensemble spread, Sec. 14

The electron reconstruction efficiency is split into three p_T ranges (< 20 , $20\text{--}75$, > 75 GeV) following the POG recommendation.

Table 38: Object-shift shape nuisances, each produced as a separate template tree.

Nuisance	Shift
<code>CMS_scale_j_2022</code>	jet energy scale (total)
<code>CMS_res_j_2022</code>	jet energy resolution
<code>CMS_scale_e_2022, CMS_res_e_2022</code>	electron scale, resolution
<code>CMS_scale_m_2022, CMS_res_m_2022</code>	muon scale, resolution

17.1 Shape uncertainties

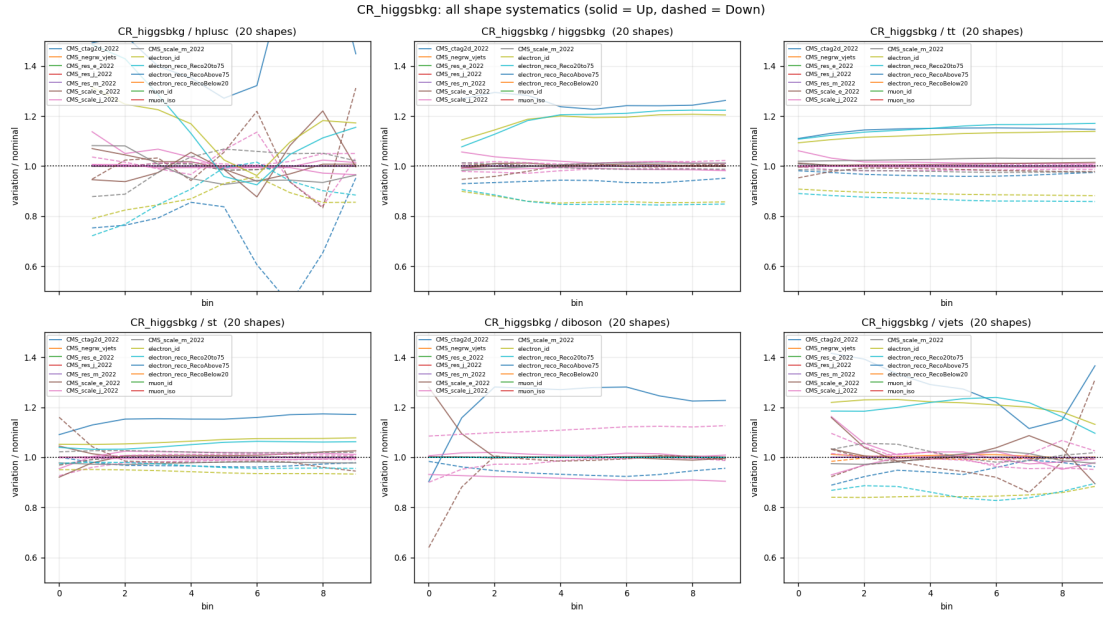


Figure 17: Shape variations of the Higgs-background template in its control region.

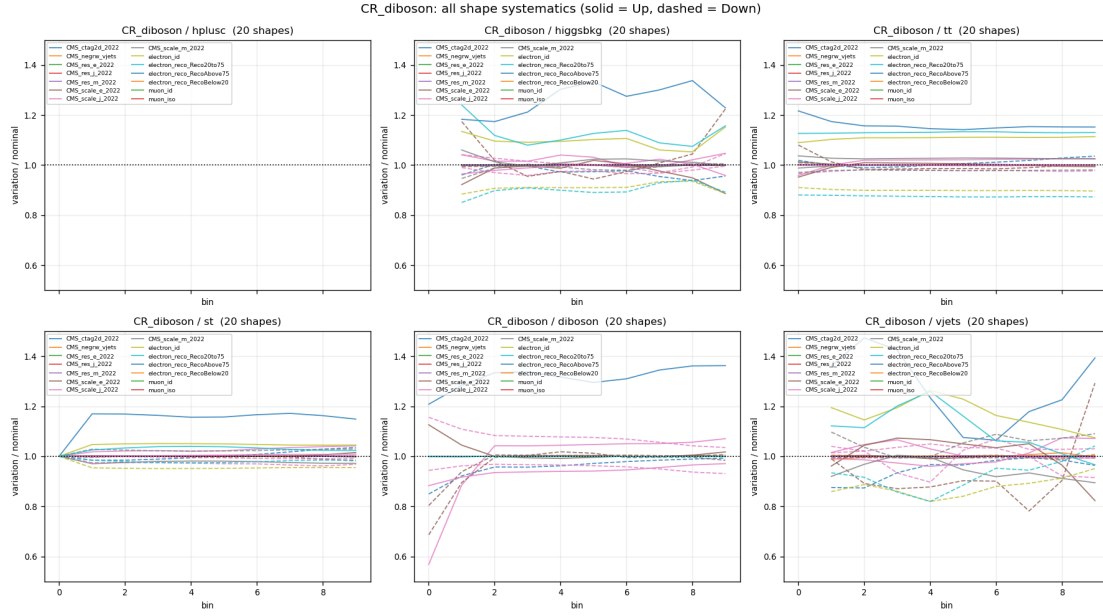


Figure 18: Shape variations of the diboson template in its control region.

17.1 Shape uncertainties

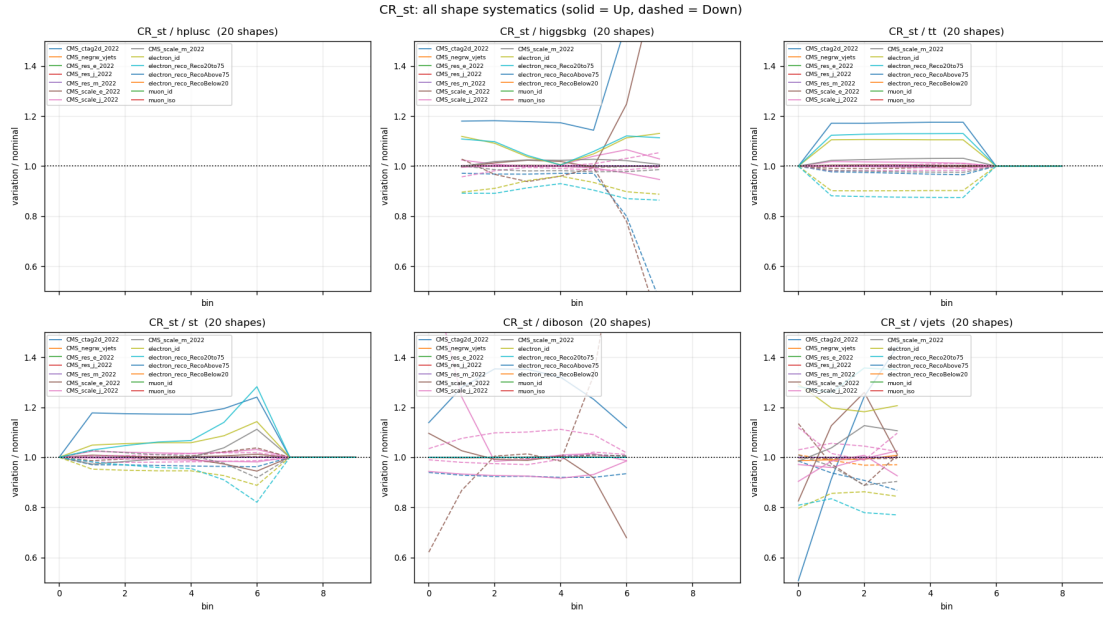


Figure 19: Shape variations of the single-top template in its control region.

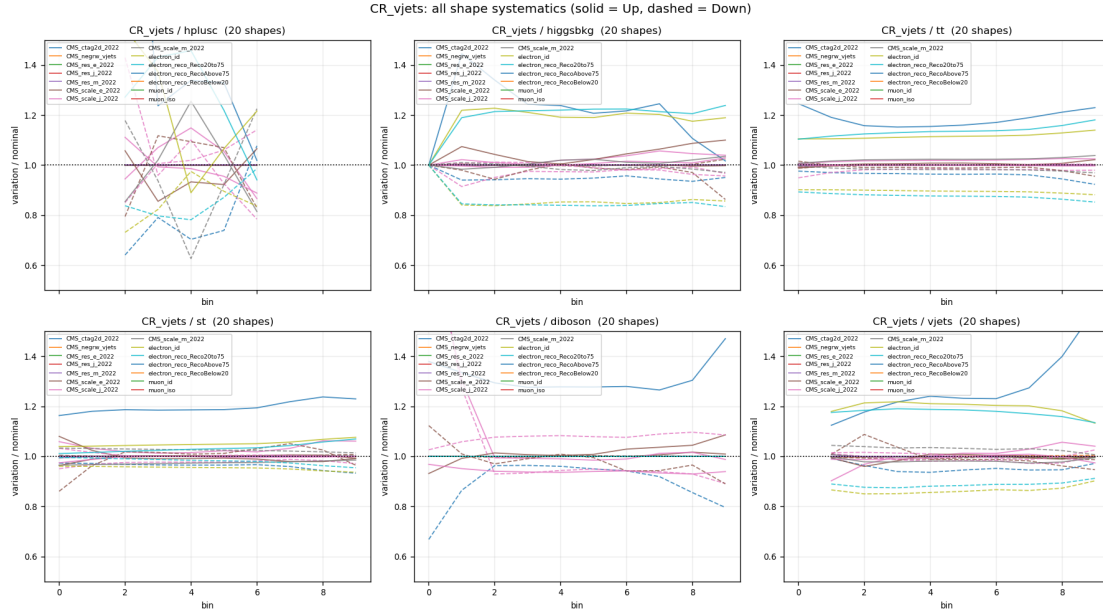


Figure 20: Shape variations of the V+jets template in the V+jets control region.

17.1 Shape uncertainties

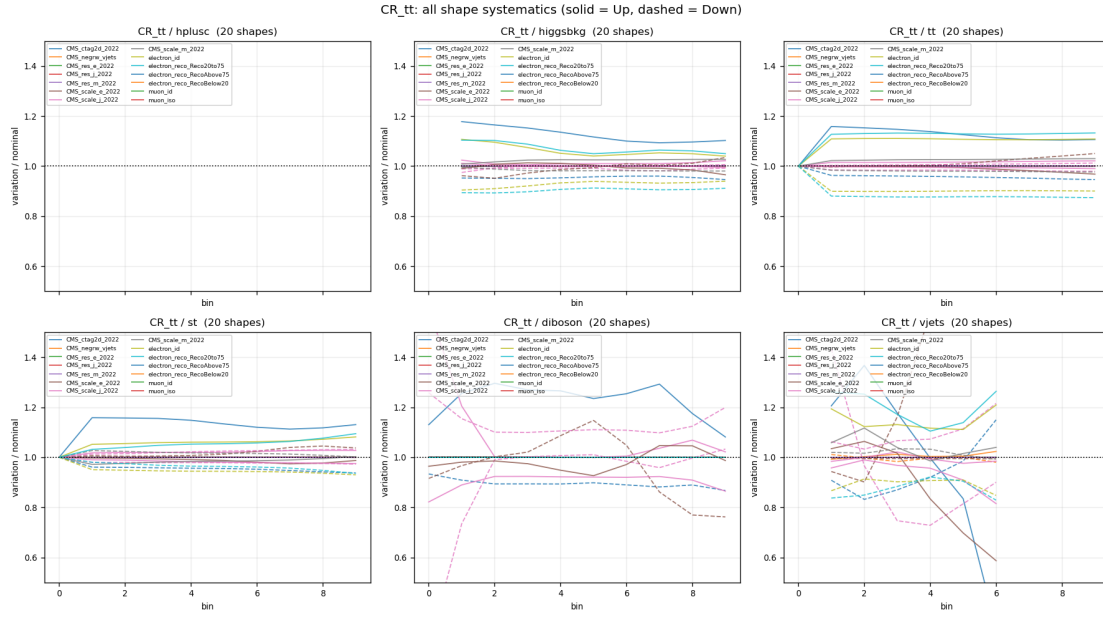


Figure 21: Shape variations of the $t\bar{t}$ template in the $t\bar{t}$ control region.

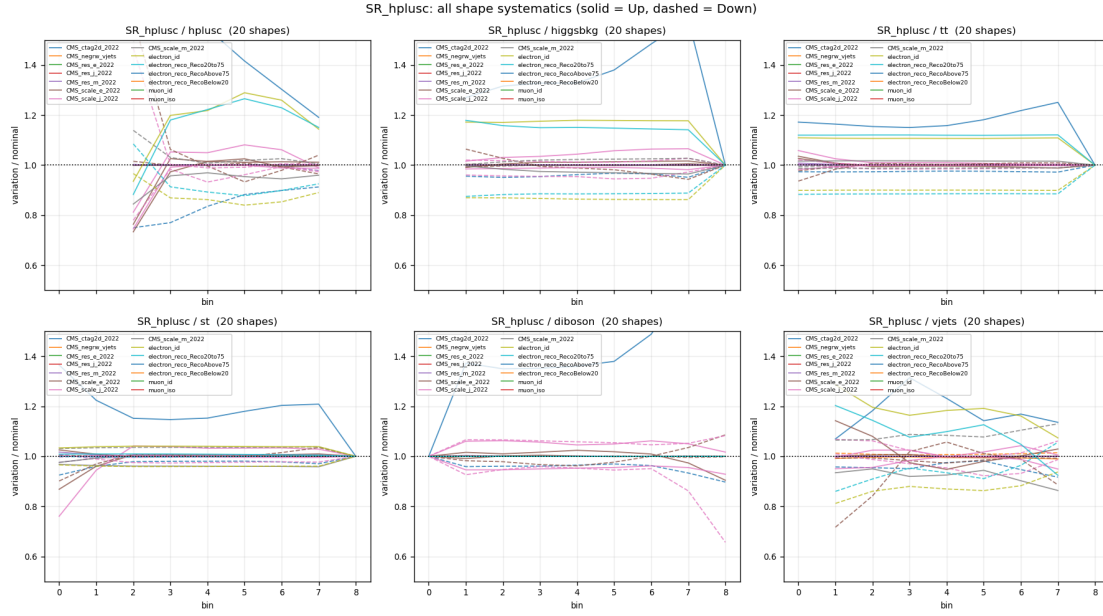


Figure 22: Shape variations of the signal template in the signal region for the principal shape nuisances.

17.2 Treatment of the $t\bar{t}$ theory uncertainties

The $t\bar{t}$ process is excluded from the parton-shower and renormalisation/factorisation scale shape nuisances above, and from the luminosity rate term. It does carry the `lhe_pdf` and `lhe_alphaS` shape nuisances, along with `pileup`, `CMS_ctag2d_2022` and the lepton efficiency terms. Its normalisation is instead left free through the `rate_tt` parameter of Sec. 16, which is unconstrained in $[0, 5]$ and shared across all six channels, and is determined by the data in the $t\bar{t}$ control region.

A free-floating rate removes the need for a normalisation uncertainty on $t\bar{t}$, but it does not cover a change in the *shape* of the $t\bar{t}$ discriminant. The PDF and α_S variations that $t\bar{t}$ does carry are the smallest of the theory sources; the parton-shower and scale variations, which are the ones that redistribute events across the MVA score and therefore across the fit categories, are absent. Since $t\bar{t}$ is 82% of the signal region, this is the leading modelling limitation of the present fit.

TODO: evaluate `ps_isr`, `ps_fsr` and the three scale variations on $t\bar{t}$ and add them as shape nuisances, normalised to leave the rate free so that they do not double-count `rate_tt`. Until this is done the quoted limit assumes the $t\bar{t}$ discriminant shape is known to better than the level these variations would produce.

17.3 Rate uncertainties

Table 39: Rate ($\ln N$) nuisances.

Nuisance	Value	Processes
<code>lumi_13p6TeV</code>	1.014	all except $t\bar{t}$
<code>xsec_st</code>	0.9873 / 1.0167	single top
<code>xsec_diboson</code>	1.037	diboson
<code>xsec_vjets</code>	1.027	V+jets
<code>xsec_higgsbkg</code>	1.05	higgsbkg
<code>BR_HtoWW</code> , <code>BR_Htautau</code>	1.01	hplusc, higgsbkg
TODO: justify <code>BR_Htautau</code> on the signal, which is $H \rightarrow WW \rightarrow 2\ell 2\nu$. If $\tau \rightarrow \ell$ contamination enters		
<code>xsec_hplusc_PDF</code>	1.06	hplusc
<code>xsec_hplusc_4FS_5FS</code>	1.30	hplusc
<code>flavor_composition_ggH</code>	1.066	higgsbkg

17.4 Top-quark p_T reweighting

The $t\bar{t}$ sample is reweighted using the theory-based (NNLO/NLO) parameterisation with the Run 3 rescaling,

$$\text{SF}_{\text{Run2}}(p_T) = 0.103 e^{-0.0118 p_T} - 1.34 \times 10^{-4} p_T + 0.973, \quad (17.1)$$

$$\text{SF}(p_T) = (0.991 + 7.5 \times 10^{-5} p_T) \text{SF}_{\text{Run2}}(p_T), \quad (17.2)$$

$$w = \sqrt{\prod_{t,\bar{t}} \text{SF}(p_T)}, \quad (17.3)$$

applied to the nominal prediction. The variation is down = 1.0, corresponding to no correction, and is symmetric about the nominal. No cap is applied at high p_T . The functional form is well behaved over the range encountered: the weight is monotonically decreasing, never negative, and takes the values 0.89 at $p_T = 1$ TeV and 0.85 at 1.5 TeV.

17.5 Higgs heavy-flavour composition

ggH constitutes only 13.1% of the merged `higgsbkg` group (VBF 29.0%, $ggZH$ 23.3%, ZH 21.0%, WH 9.1%, with the remaining 4.5% from $t\bar{t}H$ and $H + b$), so a flat $\ln N$ applied to the group either over-penalises the remaining 87% or must be diluted to an average, and in neither case produces a shape effect. The uncertainty is instead implemented as a per-event weight keyed on generator-jet flavour:

$$n_{\text{HF}} = \sum_{\text{genJets}} \mathbb{K} [|\text{hadronFlavour}| = 4, p_T > 25 \text{ GeV}, |\eta| < 2.5], \quad (17.4)$$

$$w_{\text{up/down}} = \begin{cases} 1.5 / 0.5 & n_{\text{HF}} > 0 \\ 1.0 & \text{otherwise} \end{cases} \quad (17.5)$$

corresponding to the 50% uncertainty on the ggH heavy-flavour composition. Because the weight is keyed on generator-level flavour, the grouping of processes becomes irrelevant.

TODO: the $\pm 50\%$ figure is a statement about the ggH heavy-flavour composition specifically, but the weight as written applies to any `higgsbkg` event with a generator-level c-jet, including VBF, ZH , WH and $ggZH$, where the composition is known better. This over-covers. Either restrict the weight to ggH or justify the common size. Clarify also the relation between this weight and the `flavor_composition_ggH` rate term of Table 39: they share a name but one is a shape weight and the other a 1.066 $\ln N$, and only one should be in the card at a time.

17.6 Missing transverse momentum

For Run 3 PuppimET the unclustered-energy shift is taken directly from the NanoAOD branches `ptUnclusteredUp/Down` and `phiUnclusteredUp/Down`.

18 Results

18.1 Expected limit

All results in this section are for the **2022postEE campaign only**, at 26.7 fb^{-1} , and the analysis is **blind**: the observation is set to the Asimov expectation and no signal-region data are used. The expected upper limit is obtained with Combine using the asymptotic CL_s method, with the signal-strength parameter r normalised to the SM $H + c$ prediction (Sec. 10). Limits are quoted on the dimensionless r .

TODO: state the plan for the remaining Run 3 campaigns (2022preEE, 2023preBPix, 2023postBPix), including whether the MVA is retrained per era or the 2022postEE training is applied throughout, and give the unblinding criteria.

Table 40: Expected upper limit at 95% CL on the $H + c$ signal strength, 2022postEE, 26.7 fb^{-1} . **TODO:** add the expected $\pm 1\sigma$ and $\pm 2\sigma$ bands, which Combine reports alongside the median.

Configuration	Limit
Full	1034
Statistical only	641

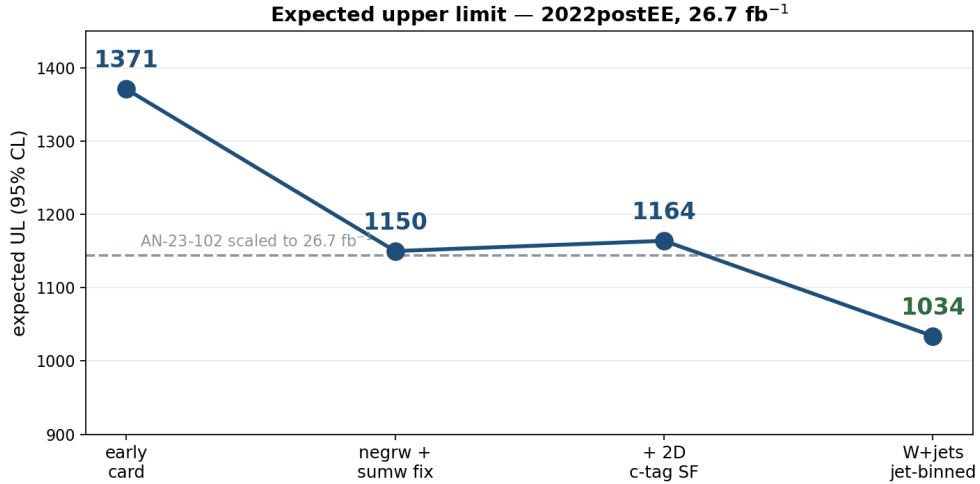


Figure 23: Development of the expected upper limit through the successive corrections described in the text.

Freezing all constrained nuisances moves the limit from 1034 to 641. Upper limits do not decompose additively, so this ratio is not a partition of the limit into statistical and systematic components and no such percentages are quoted here. The statement it does support is the comparison itself: systematic uncertainties inflate the expected limit

by a factor $1034/641 = 1.61$, and an analysis with the same data and no systematic uncertainties would reach 641.

18.2 Development of the limit

Table 41: Expected limit after each analysis change. Values before the normalisation and smoothing fixes are superseded.

Configuration	Limit	Comment
Early card	1371	superseded
+ negative-weight reweighting, <code>sumw_records</code> normalisation	1150	
+ two-dimensional charm-tag SF	1164	additional nuisance, Sec. 13
+ jet-binned W +jets	1034	Sec. 15

Two configuration changes supersede all earlier numbers: the normalisation was moved to the self-contained `sumw_records` (the previous sidecar being stale for the signal, raising the signal template by 18.4% and V +jets by 6.0%), and LOWESS template smoothing was disabled, since the negative-weight reweighting already removes the Monte Carlo statistical variance at source and the two treatments were being applied together to the V +jets templates.

18.3 Impact of individual nuisances

Each nuisance is frozen in turn, with the change in the expected limit giving its impact.

Table 42: Change in the expected limit when each source is frozen.

Frozen	Limit	Δ	% of 1034
All constrained nuisances	641	393	38.0
Theory shapes (scale + PS + α_S)	883	151	14.6
<code>xsec_hplusc_4FS_5FS</code>	921	113	10.9
<code>scalevar_muF</code>	944	90	8.7
<code>autoMCStats</code> (all)	956	78	7.5
<code>CMS_ctag2d_2022</code>	964	70	6.8
<code>rate_tt</code>	1011	23	2.2
Jet energy scale	1023	11	1.1

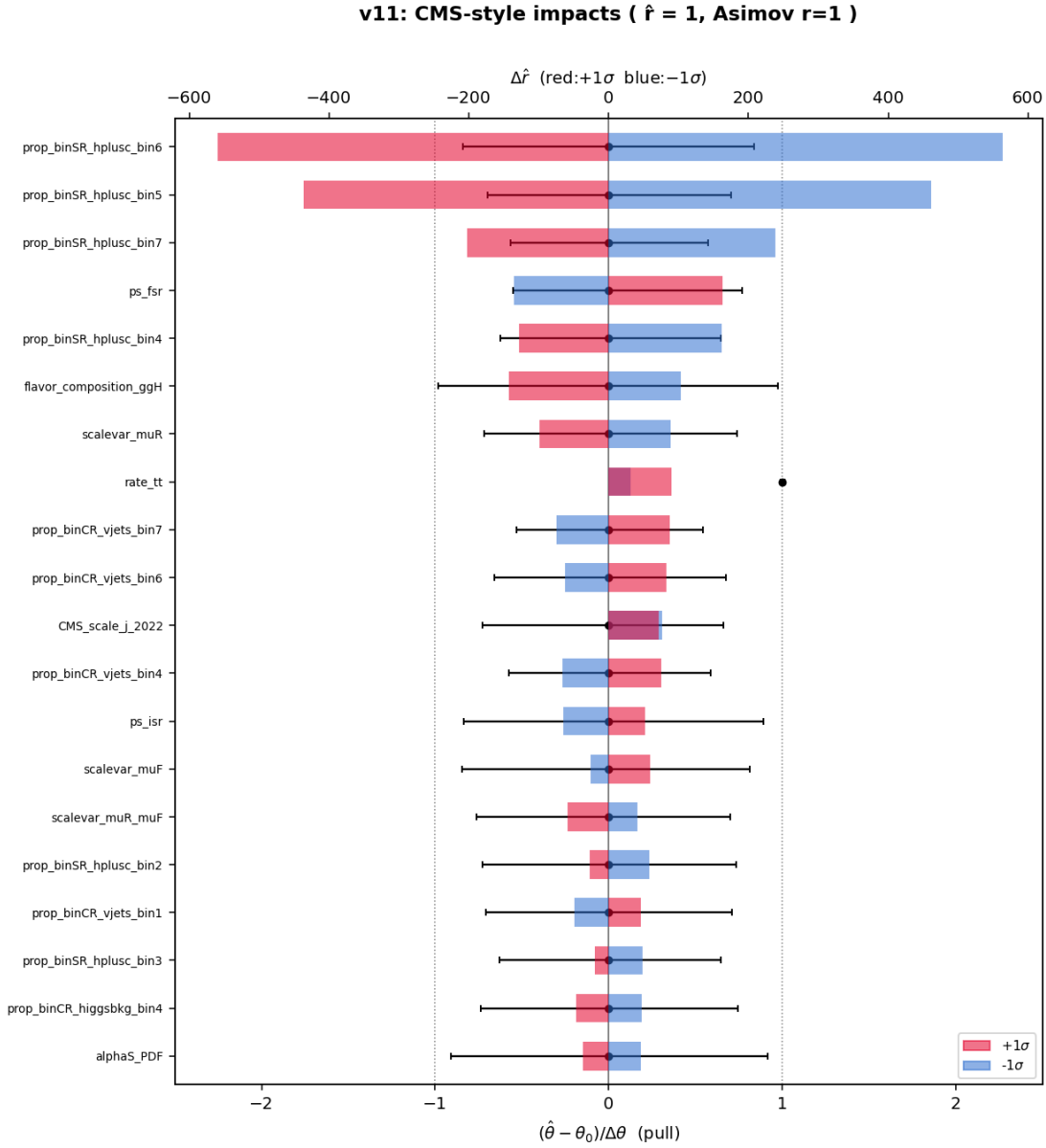


Figure 24: Impacts of the leading nuisance parameters on the signal strength, from an Asimov fit. The header value of $\hat{r}$ is a rounding artefact of the plotting script; the underlying fit returns $\hat{r} = 1.00$.

These entries overlap and are not additive: `scalevar_muF` is contained in the theory group, and freezing one nuisance allows the others to re-profile. The table should be read as a ranking rather than a decomposition.

Following the W +jets sample replacement, `autoMCStats` is no longer the leading term. The largest single contribution is now the signal-theory uncertainty `xsec_hplusc_4FS_5FS`, which is presently a placeholder value of 1.30 and requires re-derivation at $\sqrt{s} =$

13.6 TeV.

TODO: `xsec_hplusc_4FS_5FS` is the largest single nuisance impact in the fit (113, or 10.9%) and its value of 1.30 is a placeholder. It needs a derivation at 13.6 TeV with a stated method and a timeline. Note also that it is applied as a rate `lnN` and therefore carries no shape: the four- and five-flavour schemes differ specifically in the charm-jet kinematics, so a shape component should be evaluated rather than assumed absent.

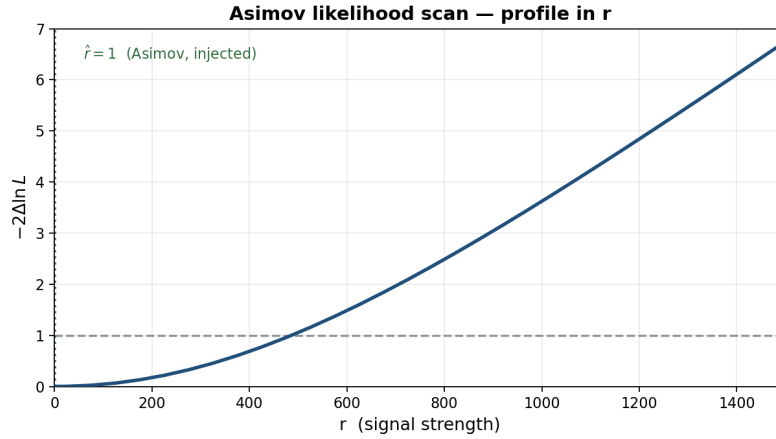


Figure 25: Profile likelihood scan of the signal strength. The minimum sits at the injected value and the $-2\Delta \ln L = 3.84$ crossing is close to the quoted CL_s limit, as expected when the two hypotheses are well separated.

18.4 Fit diagnostics

On the Asimov dataset the fit returns $\hat{r} = 1.00$. All Gaussian nuisances are found at zero pull, which is a property of the Asimov construction rather than a result: the informative content of Fig. 26 is the post-fit *widths*, since a nuisance constrained appreciably below unit width is being determined by the templates rather than by its external prior.

TODO: quote the most strongly constrained nuisances and their post-fit widths, and justify any constraint well below unit width, in particular for the theory shapes and `CMS_ctag2d_2022`. An over-constrained systematic usually indicates either a template statistics artefact or a nuisance that the fit is using to absorb a genuine shape difference. Two parameters are not Gaussian and should not be interpreted as pulls: `rate_tt`, the free-floating rate parameter, which sits at 1.000 ± 0.017 constrained by `CR_tt`; and one `prop_bin` parameter in a bin with $n_{\text{eff}} = 2.1$, below the `autoMCStats` threshold, which therefore receives an individual Poisson term at its nominal scale.

The profile likelihood in r has its minimum at the injected value, with the $-2\Delta \ln L = 1$ crossing at $r \approx 525$ and the $-2\Delta \ln L = 3.84$ crossing at $r \approx 1075$, the latter close to the quoted CL_s limit of 1034 as expected when the two hypotheses are well separated.

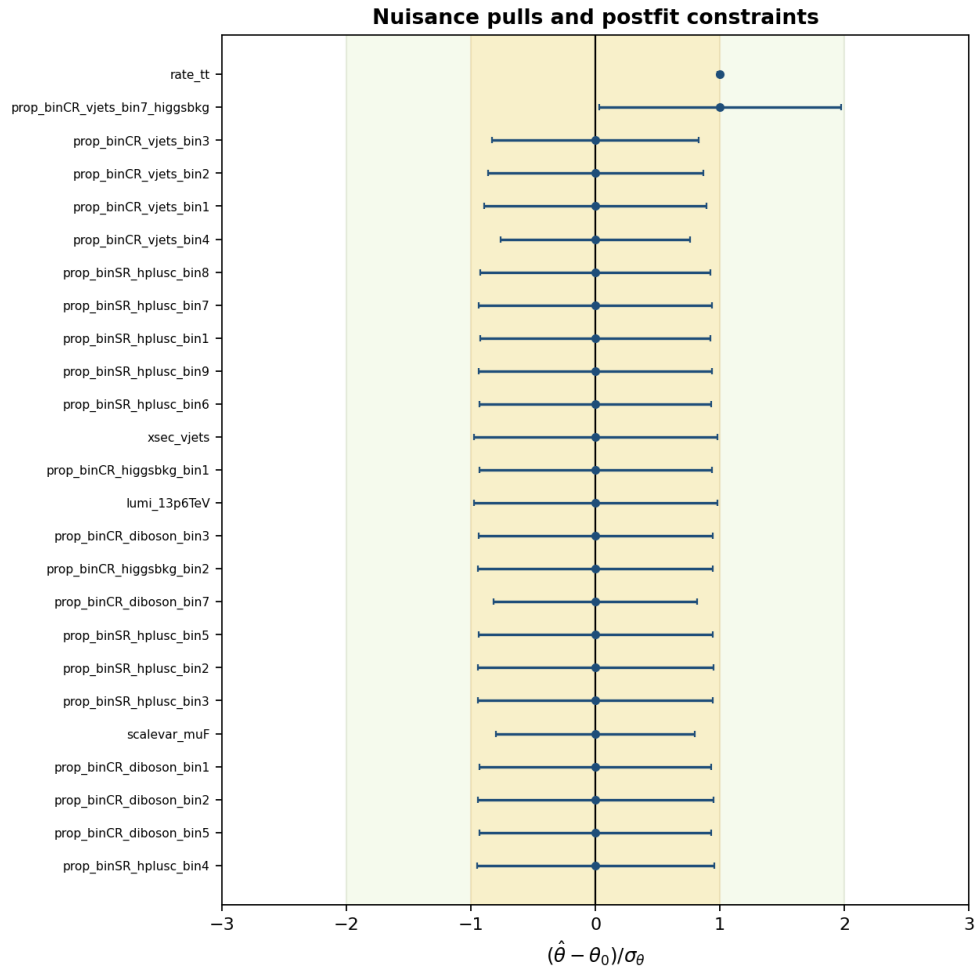


Figure 26: Nuisance-parameter pulls and constraints from the Asimov fit.

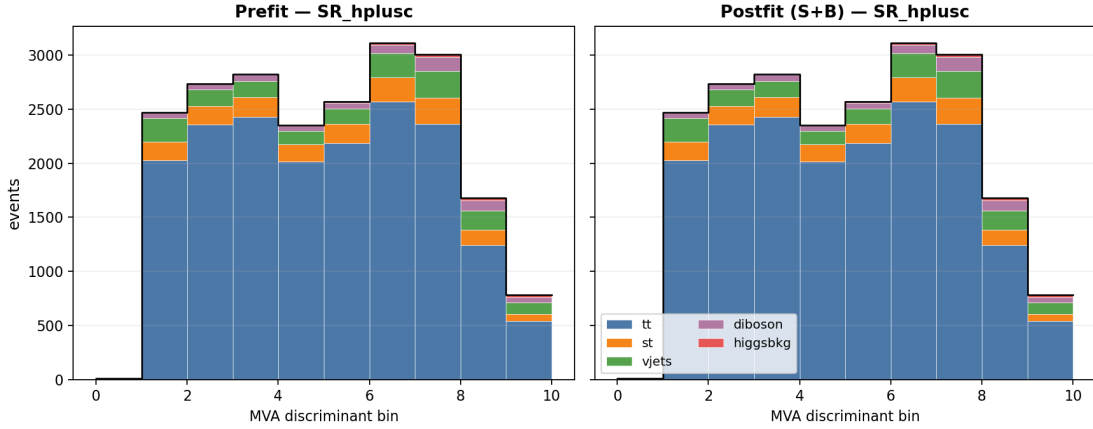


Figure 27: Prefit and postfit signal-region distributions. The two are nearly identical because the fit is performed on an Asimov dataset built from the prefit model, so no pull is possible by construction; the comparison verifies that the fit does not distort the templates rather than testing the model.

18.5 Comparison with the Run 2 analysis

AN-23-102 [3] quotes an expected upper limit of 503 at 138 fb^{-1} . A naive statistics-limited extrapolation to the luminosity used here gives

$$503 \times \sqrt{138/26.7} = 1144, \quad (18.1)$$

against 1034 obtained in this analysis. This comparison should be read with three caveats, and is not yet a like-for-like statement.

First, $\sqrt{\mathcal{L}}$ scaling is valid only for a statistics-limited result. The Run 2 analysis is itself only 73.8% statistical, so the systematic part of its uncertainty does not shrink with luminosity and the correctly extrapolated value is larger than 1144. Scaling the statistical and systematic parts separately would give a range rather than a single number, and the naive figure quoted above therefore flatters the present analysis.

Second, the two analyses are at different centre-of-mass energies, 13 TeV and 13.6 TeV. The limit is quoted on the dimensionless signal strength r , so the comparison is meaningful only if both are normalised to the same reference cross section times branching fraction.

TODO: confirm that the Run 2 limit of 503 and the limit quoted here are normalised to the same $\sigma \times B$ convention, and state that convention explicitly. If the references differ, rescale before comparing. Also confirm whether 503 is the $e\mu$ -only result or a combination of channels, since this analysis is $e\mu$ only. Until both are settled, the comparison above is indicative and not a claim of improvement.

References

- [1]
- [2] Positive-weight reweighting of negative-weight Monte Carlo events, [arXiv:2510.16217](#).
- [3] CMS Collaboration, *Search for Higgs boson production in association with a charm quark in the $H \rightarrow WW$ decay channel*, CMS AN-23-102.
- [4] M. Kronheim and C. Palmer, *PRISM-PCA: removing negative weights from aMC@NLO samples*, CMS Analysis Note (2026).